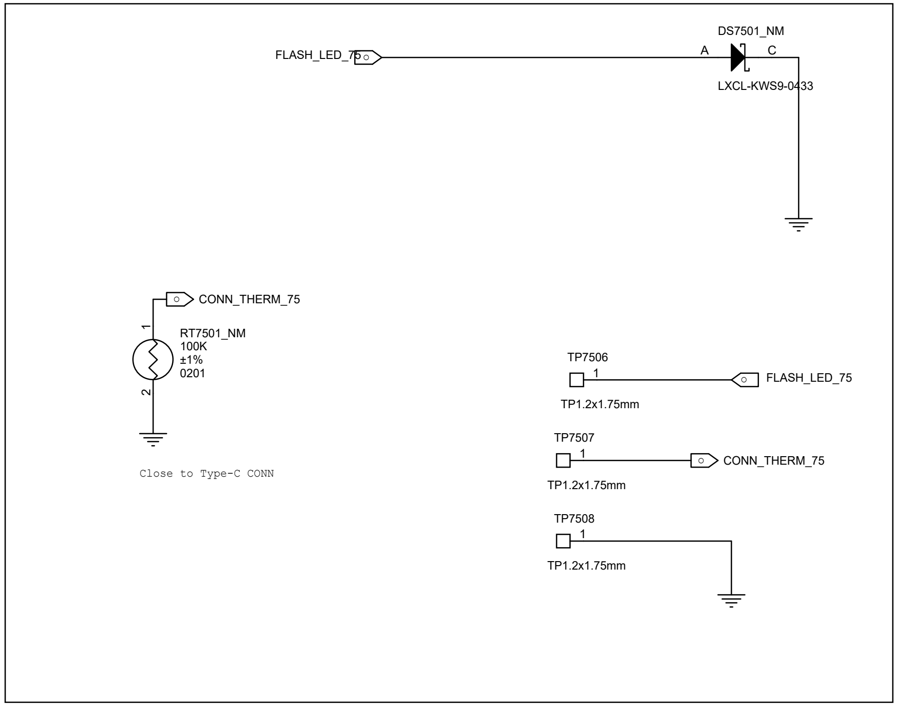
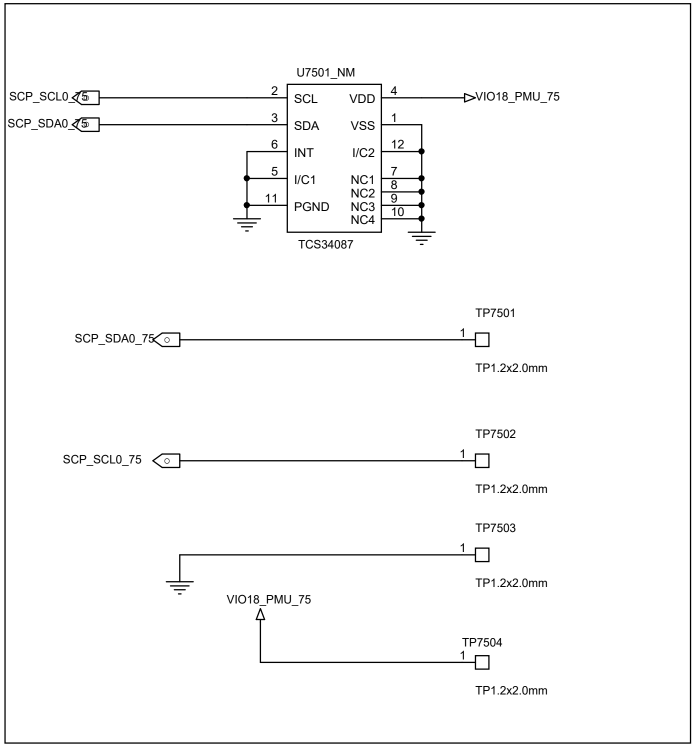
FLASH LGA



L_SENSOR LGA



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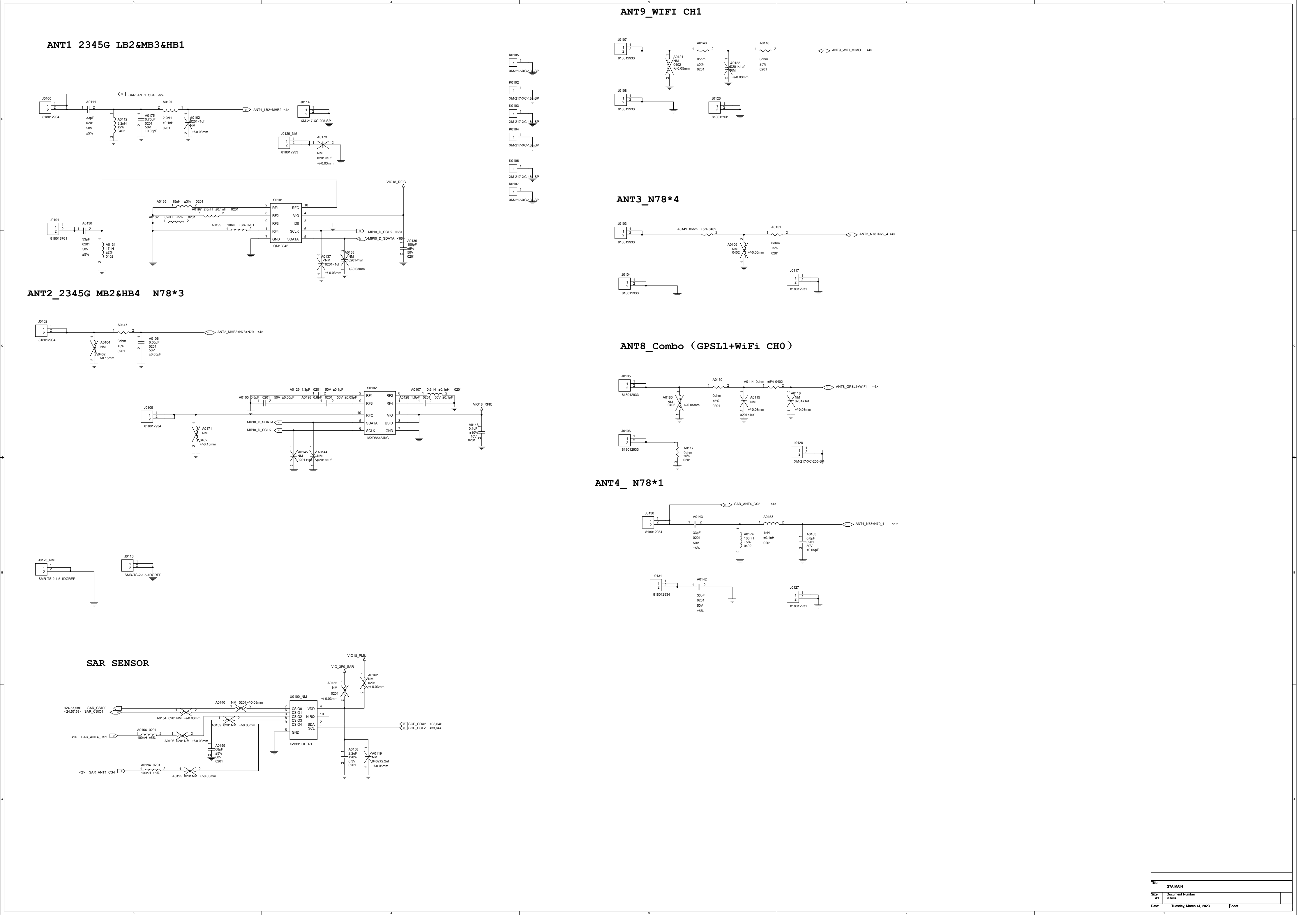
Size A3	Document Number <Doc>	Rev <RevCode>
Date: Tuesday, March 14, 2023	Sheet 1 of 1	



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MIPI0_D_SCLK <66>
MIPI0_D_SDATA <66>
QM13346
ANT2_2345G MB2&HB4 N78*3
ANT2_MHB3+N78+N79 <4>
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MIPI0_D_SCLK
MXD8548JKC
ANT9_WIFI CH1
ANT9_WIFI_MIMO <4>
ANT3_N78*4
ANT3_N78+N79_4 <4>
ANT8_Combo（GPSL1+WiFi CH0）
ANT8_GPSL1+WIFI <4>
ANT4_ N78*1
SAR_ANT4_CS2 <4>
ANT4_N78+N79_1 <4>
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VIO18_PMU
VIO_3P0_SAR
SAR_CSIO0
SAR_CSIO1
<2> SAR_ANT4_CS2
<2> SAR_ANT1_CS4
SCP_SDA2 <33,64>
SCP_SCL2 <33,64>
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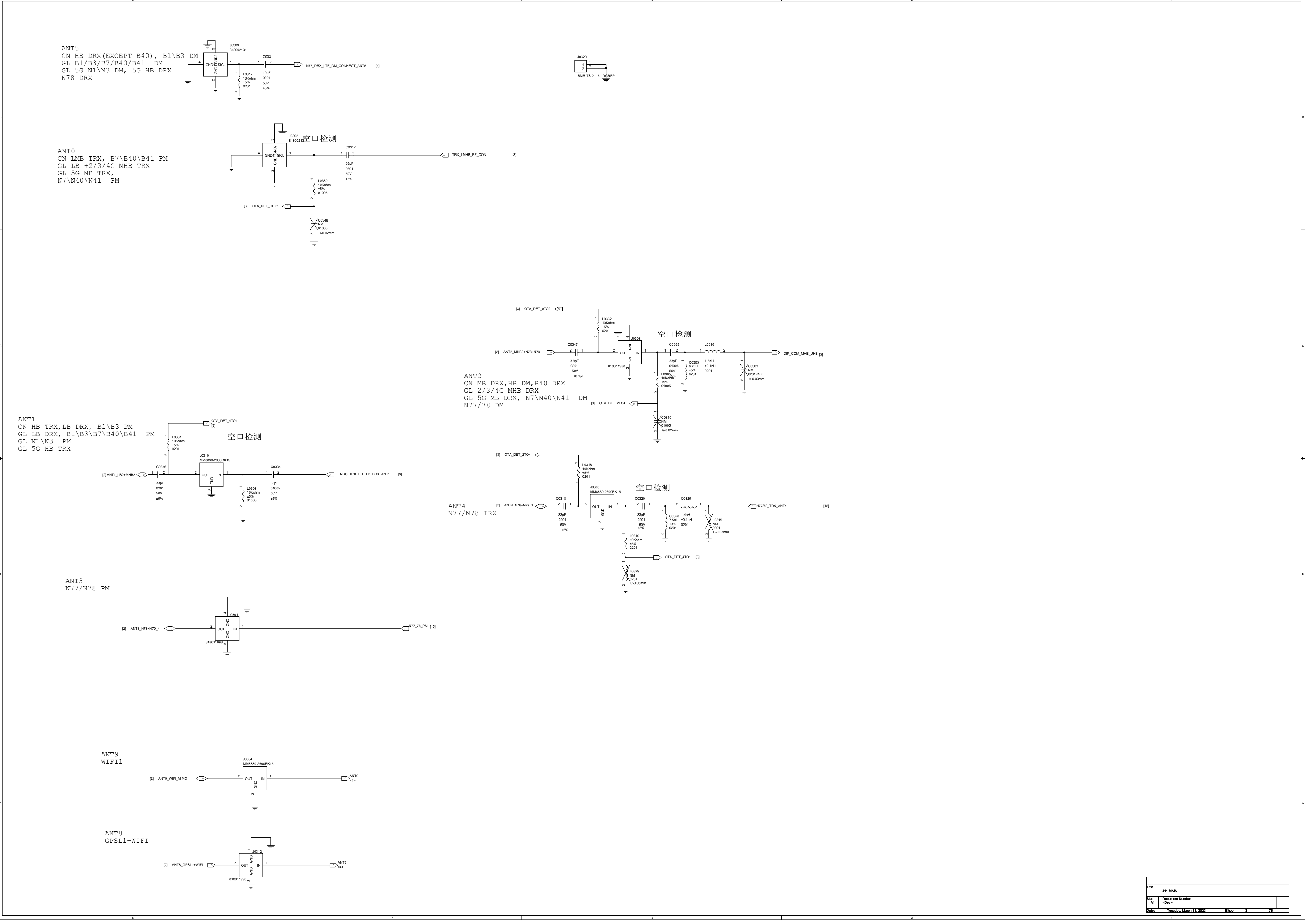
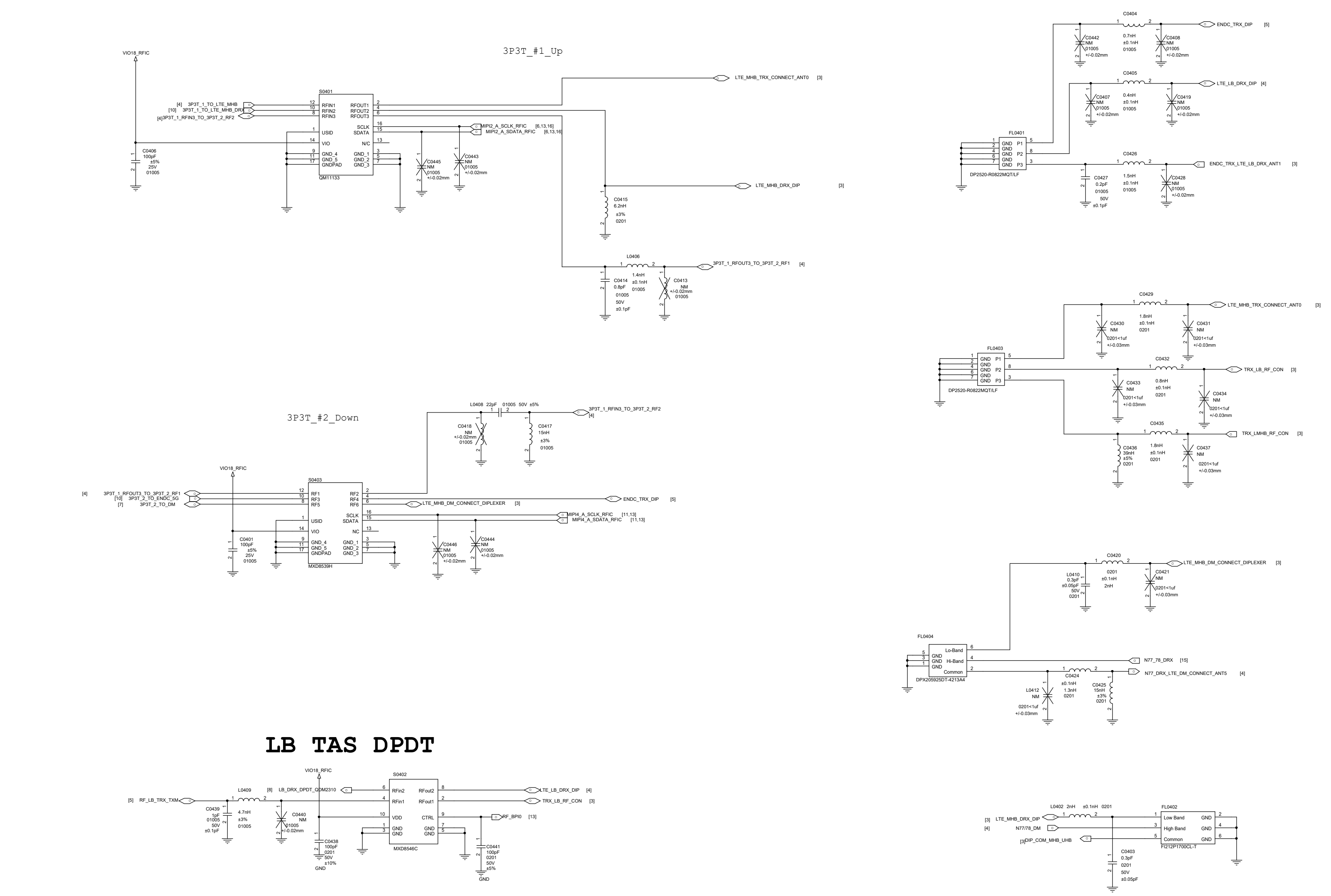
G7A MAIN
Size A1
Document Number <Doc>
Date: Tuesday, March 14, 2023
Sheet


ANT5
CN HB DRX(EXCEPT B40), B1\B3 DM
GL B1/B3/B7/B40/B41 DM
GL 5G N1\N3 DM, 5G HB DRX
N78 DRX
J0303
818002131
C0331
10pF
0201
50V
±5%
L0317
10Kohm
±5%
0201
N77_DRX_LTE_DM_CONNECT_ANT5
[4]
J0320
SMR-TS-2-1.5-1DGREP
ANT0
CN LMB TRX, B7\B40\B41 PM
GL LB +2/3/4G MHB TRX
GL 5G MB TRX,
N7\N40\N41 PM
J0302
818002131
空口检测
C0317
33pF
0201
50V
±5%
TRX_LMHB_RF_CON
[3]
L0330
10Kohm
±5%
01005
[3] OTA_DET_0TO2
C0348
NM
01005
+/-0.02mm
[3] OTA_DET_0TO2
L0332
10Kohm
±5%
0201
J0308
空口检测
C0347
3.9pF
0201
50V
±0.1pF
[2] ANT2_MHB3+N78+N79
818011998
C0335
33pF
01005
50V
L0310
1.5nH
±0.1nH
0201
C0303
8.2nH
±5%
0201
C0309
NM
0201<1uf
+/-0.03mm
DIP_COM_MHB_UHB
[3]
L0305
10Kohm
±5%
01005
ANT2
CN MB DRX,HB DM,B40 DRX
GL 2/3/4G MHB DRX
GL 5G MB DRX, N7\N40\N41 DM
N77/78 DM
[3] OTA_DET_2TO4
C0349
NM
01005
+/-0.02mm
ANT1
CN HB TRX,LB DRX, B1\B3 PM
GL LB DRX, B1\B3\B7\B40\B41 PM
GL N1\N3 PM
GL 5G HB TRX
OTA_DET_4TO1
[3]
L0331
10Kohm
±5%
0201
空口检测
J0310
MM8830-2600RK15
[2] ANT1_LB2+MHB2
C0346
33pF
0201
50V
±5%
L0306
10Kohm
±5%
01005
C0334
33pF
01005
50V
±5%
ENDC_TRX_LTE_LB_DRX_ANT1
[3]
[3] OTA_DET_2TO4
L0318
10Kohm
±5%
0201
J0305
MM8830-2600RK15
空口检测
ANT4
N77/N78 TRX
[2] ANT4_N78+N79_1
C0318
33pF
0201
50V
±5%
C0320
33pF
0201
50V
±5%
C0326
7.5nH
±3%
0201
C0325
1.4nH
±0.1nH
0201
L0315
NM
0201
+/-0.03mm
N77/78_TRX_ANT4
[15]
L0319
10Kohm
±5%
0201
OTA_DET_4TO1
[3]
L0329
NM
0201
+/-0.03mm
ANT3
N77/N78 PM
J0301
[2] ANT3_N78+N79_4
818011998
N77_78_PM
[15]
ANT9
WIFI1
J0304
MM8830-2600RK15
[2] ANT9_WIFI_MIMO
ANT9
<4>
ANT8
GPSL1+WIFI
J0312
[2] ANT8_GPSL1+WIFI
818011998
ANT8
<4>
Title
J11 MAIN
Size
A1
Document Number
<Doc>
Date: Tuesday, March 14, 2023
Sheet 3 76



MI PHONE	
Title	MT6883_5GL
Size A1	Document Number SWITCH — Rev P0
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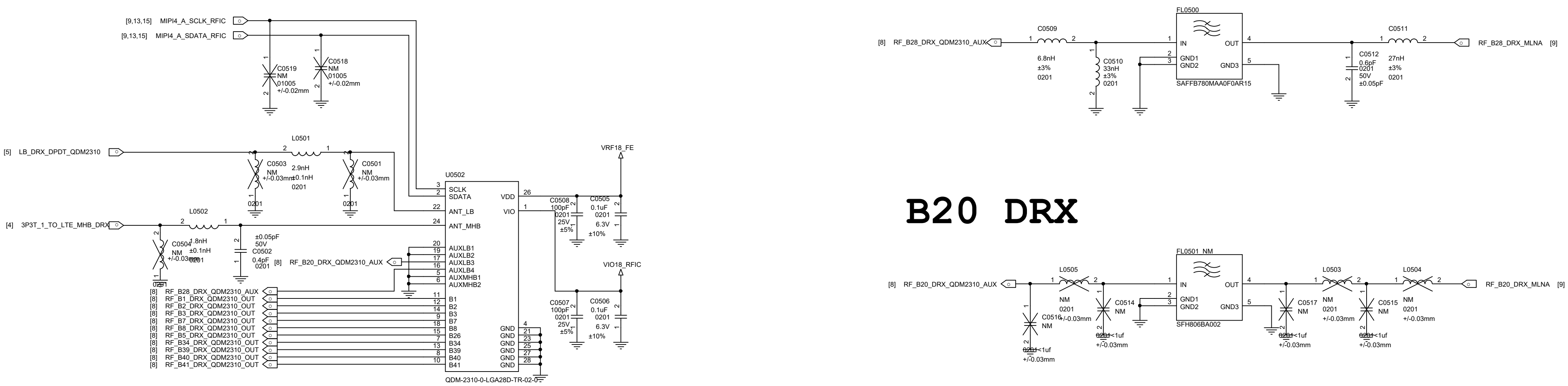
DiFEM QDM2310

B28 DRX

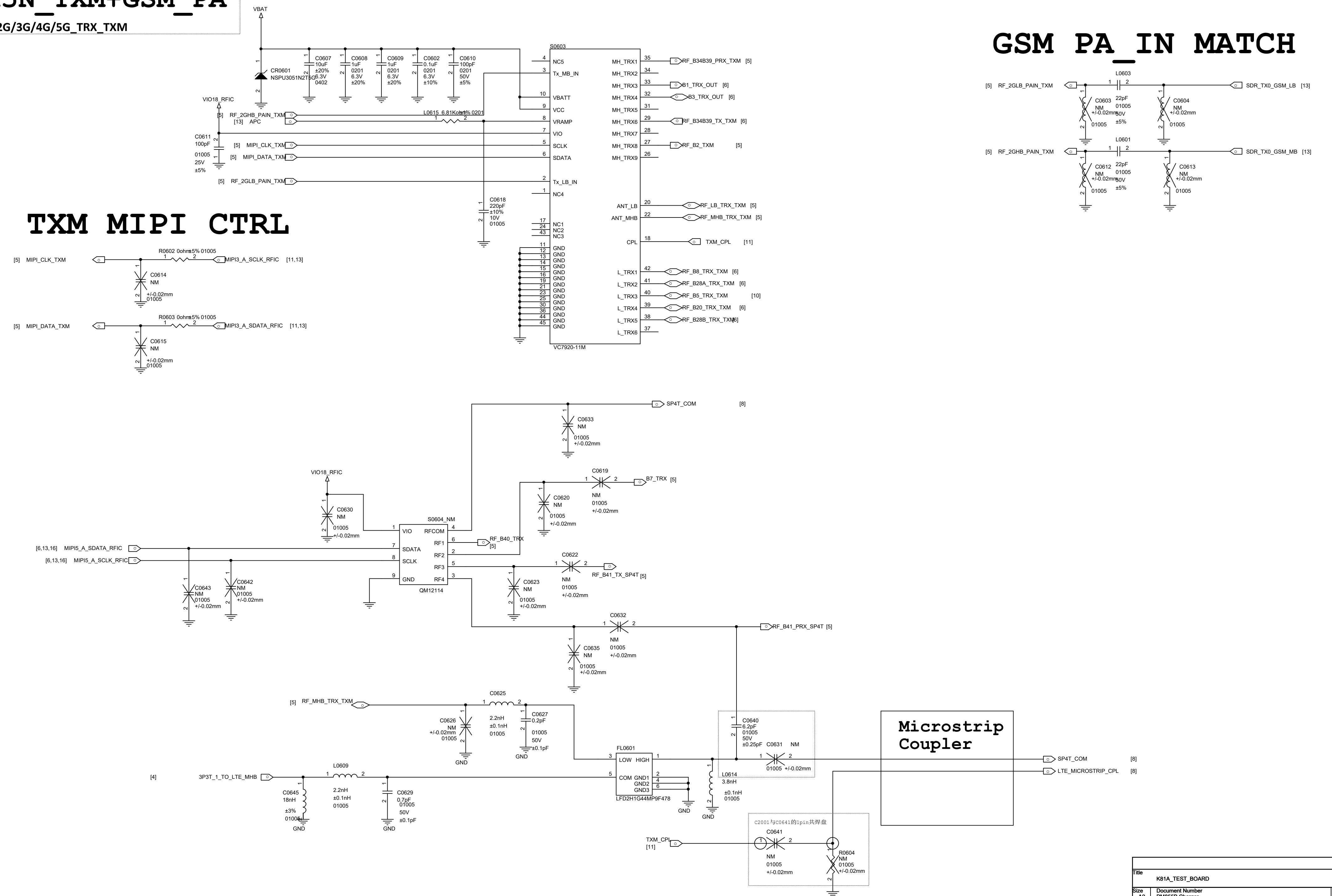
B20 DRX



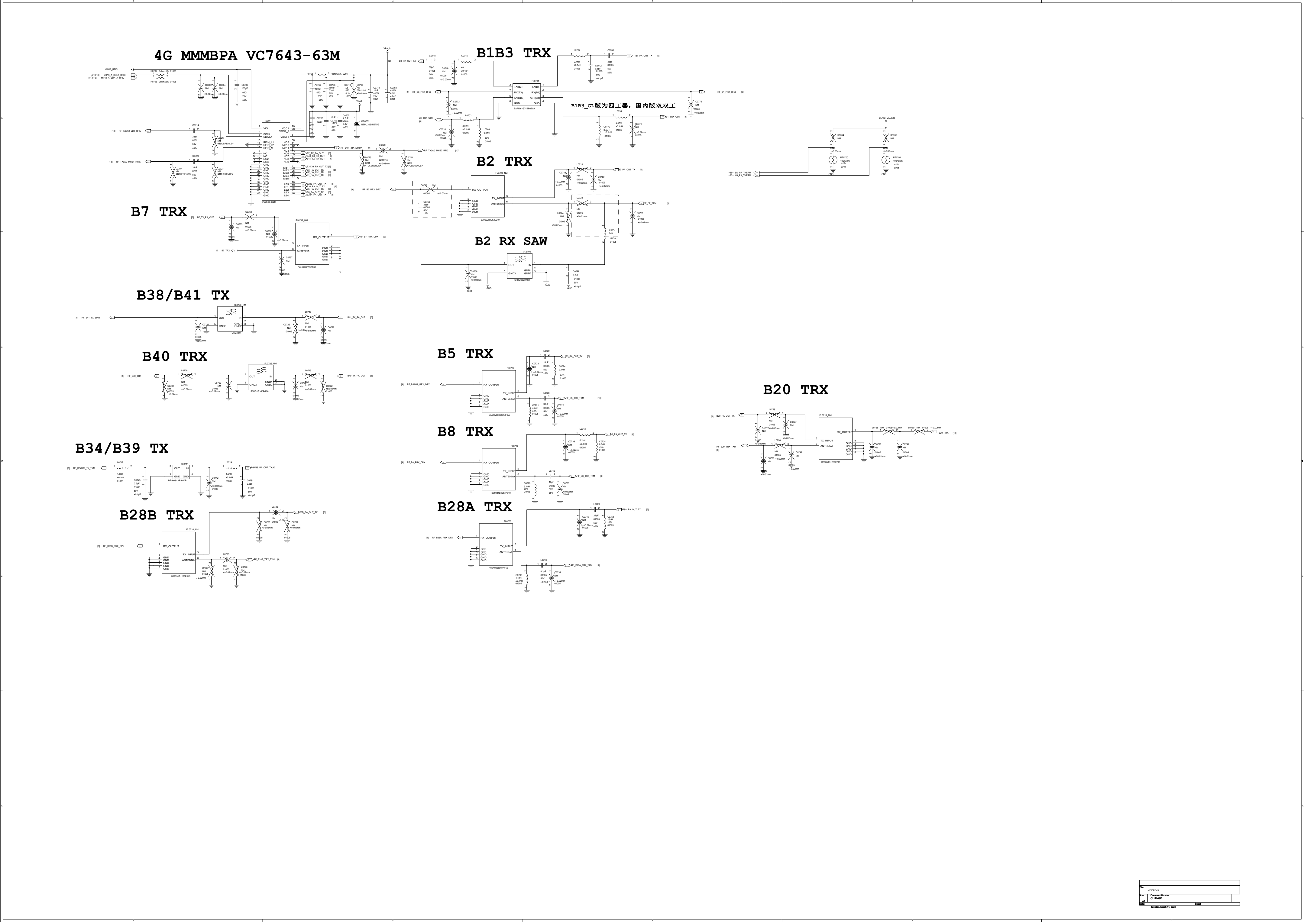
Title	MT6873_5GA
Size	Document Number
A2	MT6893_GPIO_1
Date:	Tuesday, March 14, 2023
Sheet	5 of 76

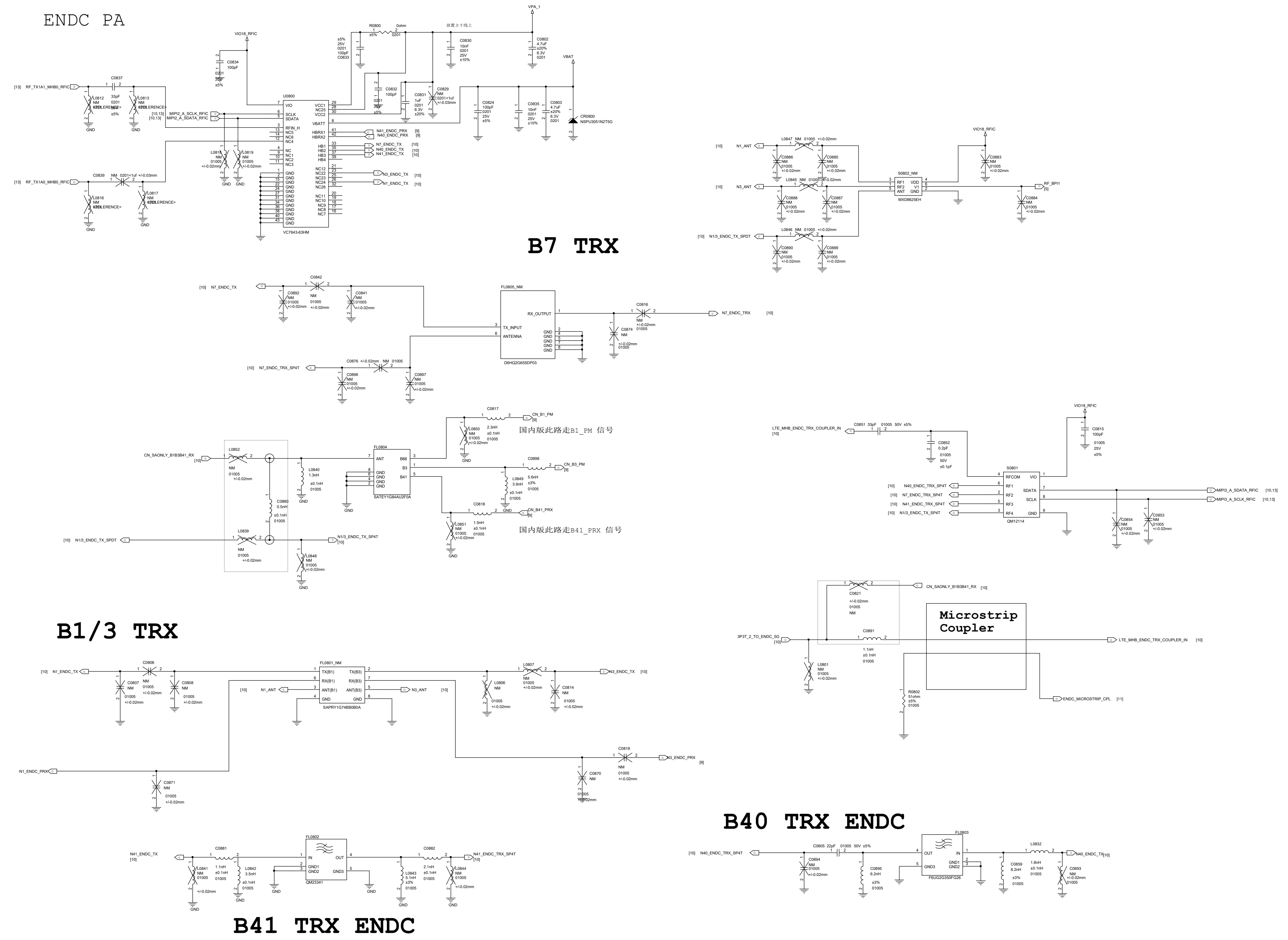
Ph5N_TXM+GSM_PA

2G/3G/4G/5G_TRX_TXM



Title	K81A_TEST_BOARD
Size A2	Document Number PM855B Charger
Date: Tuesday, March 14, 2023	Sheet 6 of 76


4G MMMBPA VC7643-63M
B1B3 TRX
B1B3_GL版为四工器，国内版双双工
B2 TRX
B2 RX SAW
B7 TRX
B38/B41 TX
B40 TRX
B5 TRX
B8 TRX
B20 TRX
B34/B39 TX
B28B TRX
B28A TRX



MI PHONE

Title: MT6883_5GL

Size	Document Number	Rev
A1	DRX	P0

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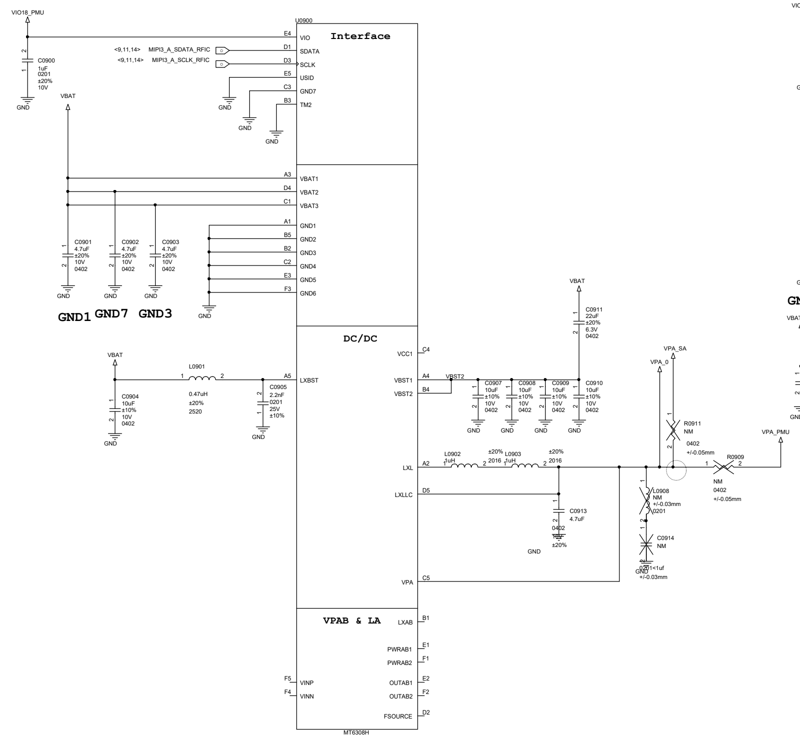
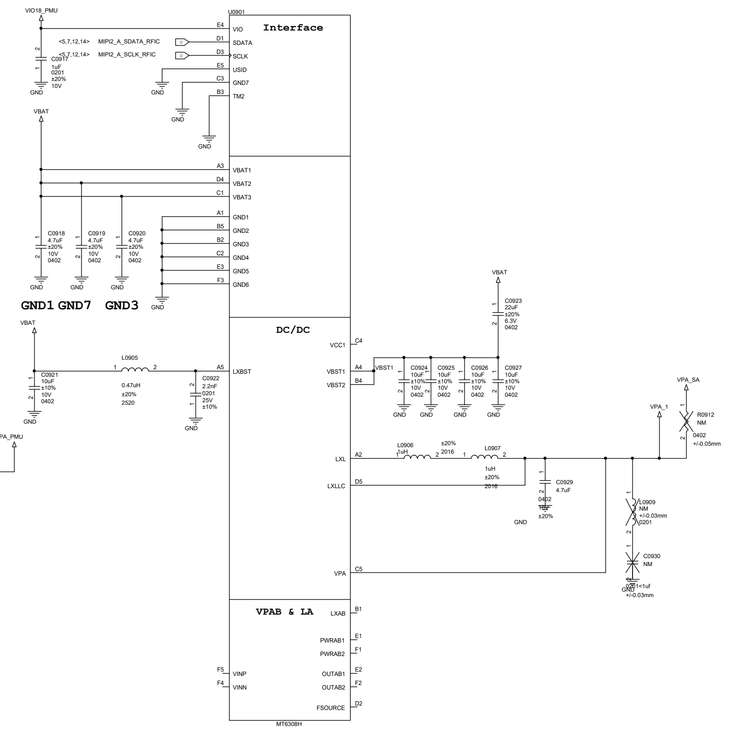
4G PA DC-DC APT ONLY



5G PA DC-DC APT ONLY



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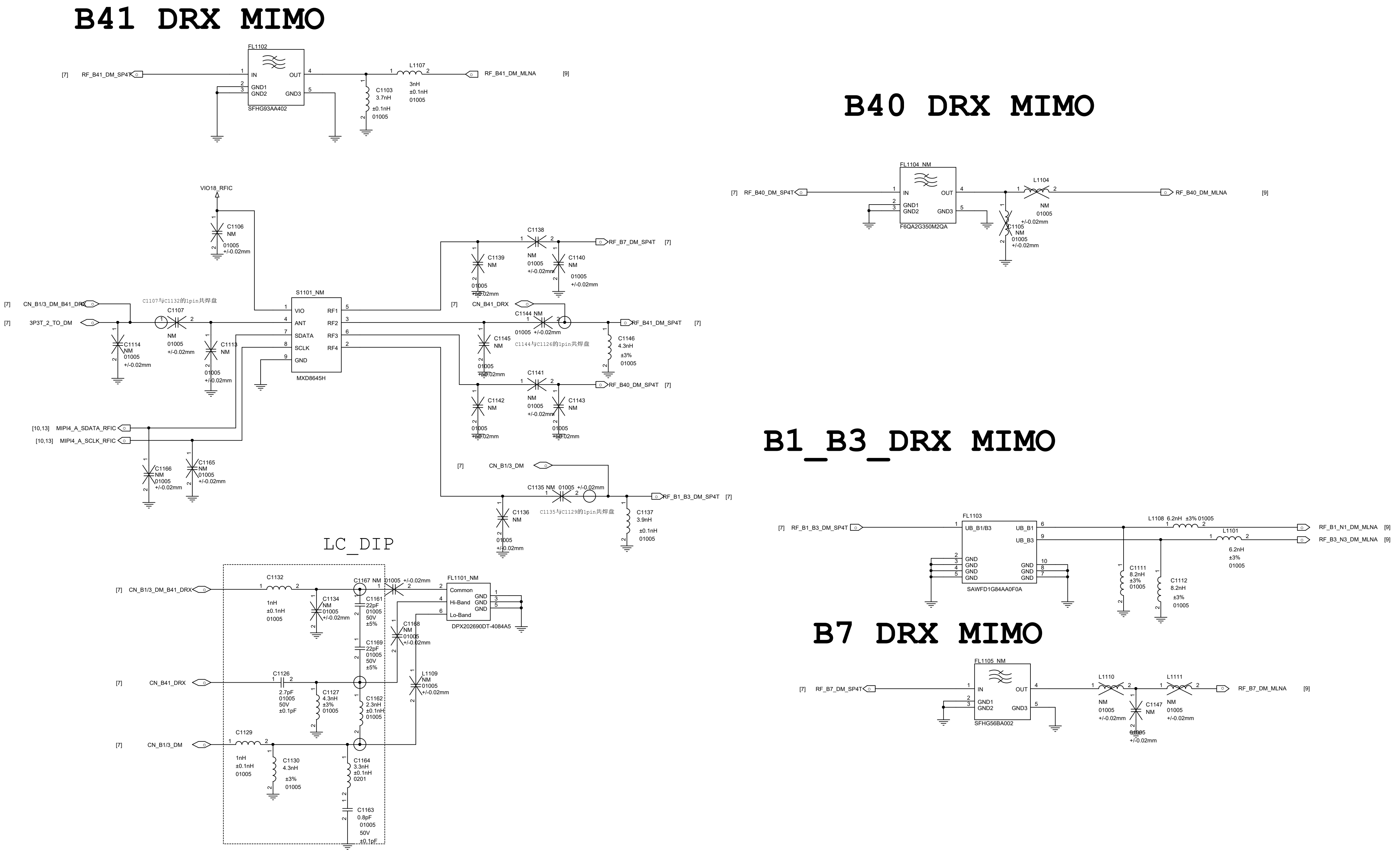
Title: MT6883_5GL		
Size: A1	Document Number: DRX_LNA	Rev: P0
Date: Tuesday, March 14, 2023	Sheet 9 of 76	



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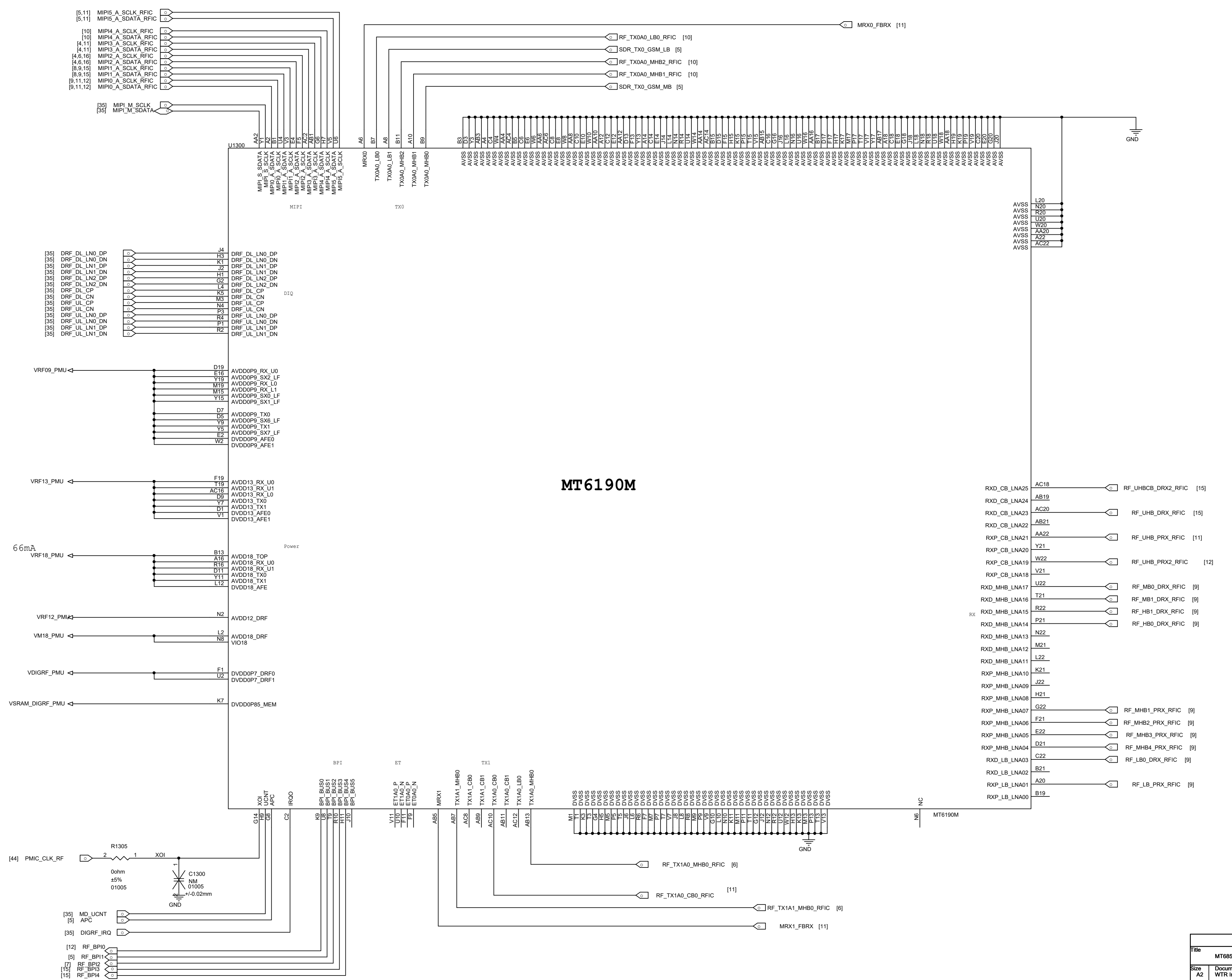
Title	MT6883_5GL	
Size A2	Document Number PA_5G_N41	Rev P0
Date: Tuesday, March 14, 2023	Sheet 11 of 76	

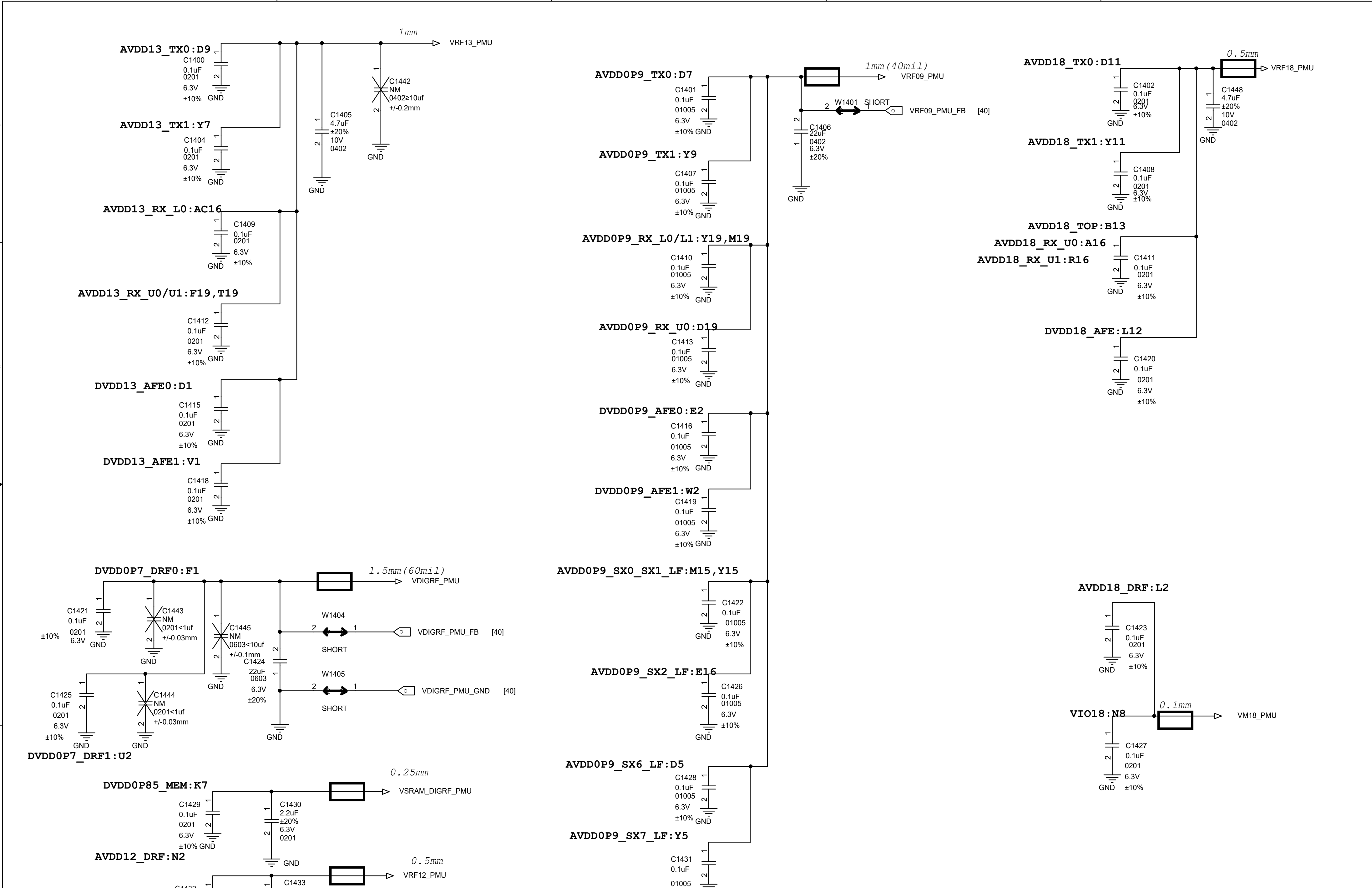


MI PHONE

Title: MT6853

Size	Document Number	Rev
A2	WTR Interface	P0

Date: Tuesday, March 14, 2023 Sheet 13 of 76



MI PHONE		
Title: MT6853		
Size: A3	Document Number: WTR PWR	Rev: P0
Date: Tuesday, March 14, 2023	Sheet 14 of 76	

PRX LNA-Bank MXD83K5A Matching

PRX HB

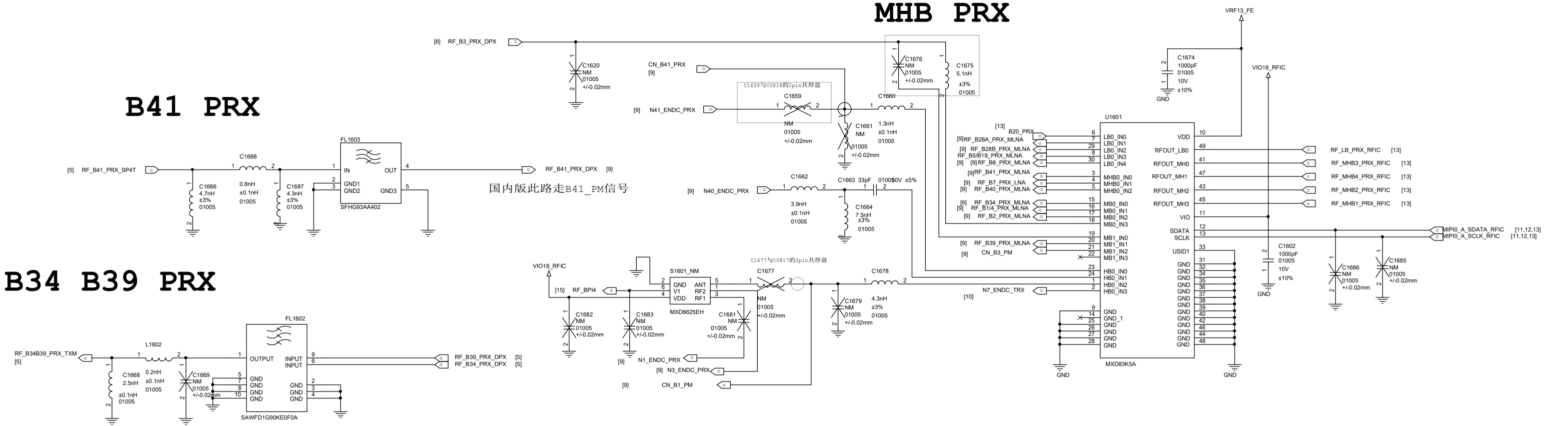
PRX MB

PRX LB

MHB PRX

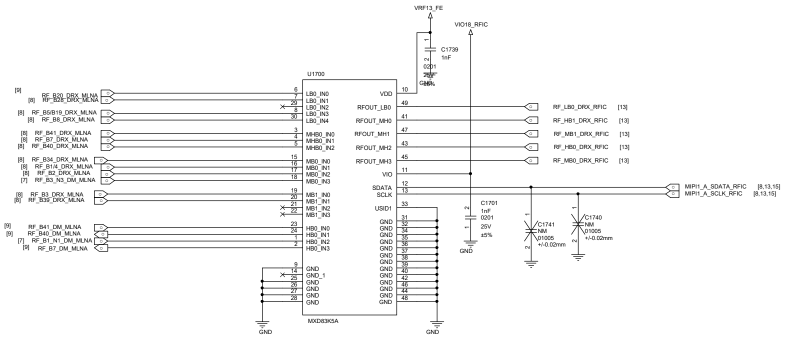
B41 PRX

B34 B39 PRX

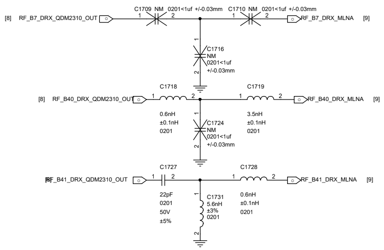


Title	
Size A1	Document Number CHANGE
Date: Tuesday, March 14, 2023	Sheet 16 of 73

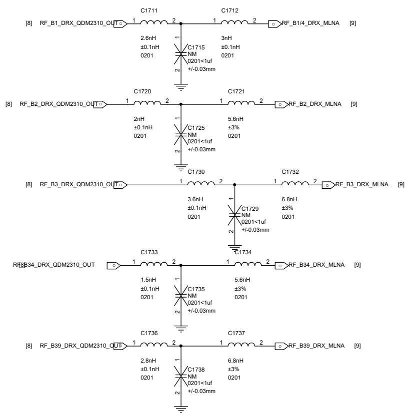
MHB DRX



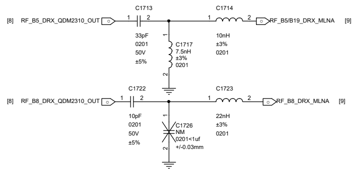
DRX HB



DRX MB



DRX LB



MI PHONE		
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Size: A1	Document Number: Reserved(for 5G-sub 6G)	Rev: P0
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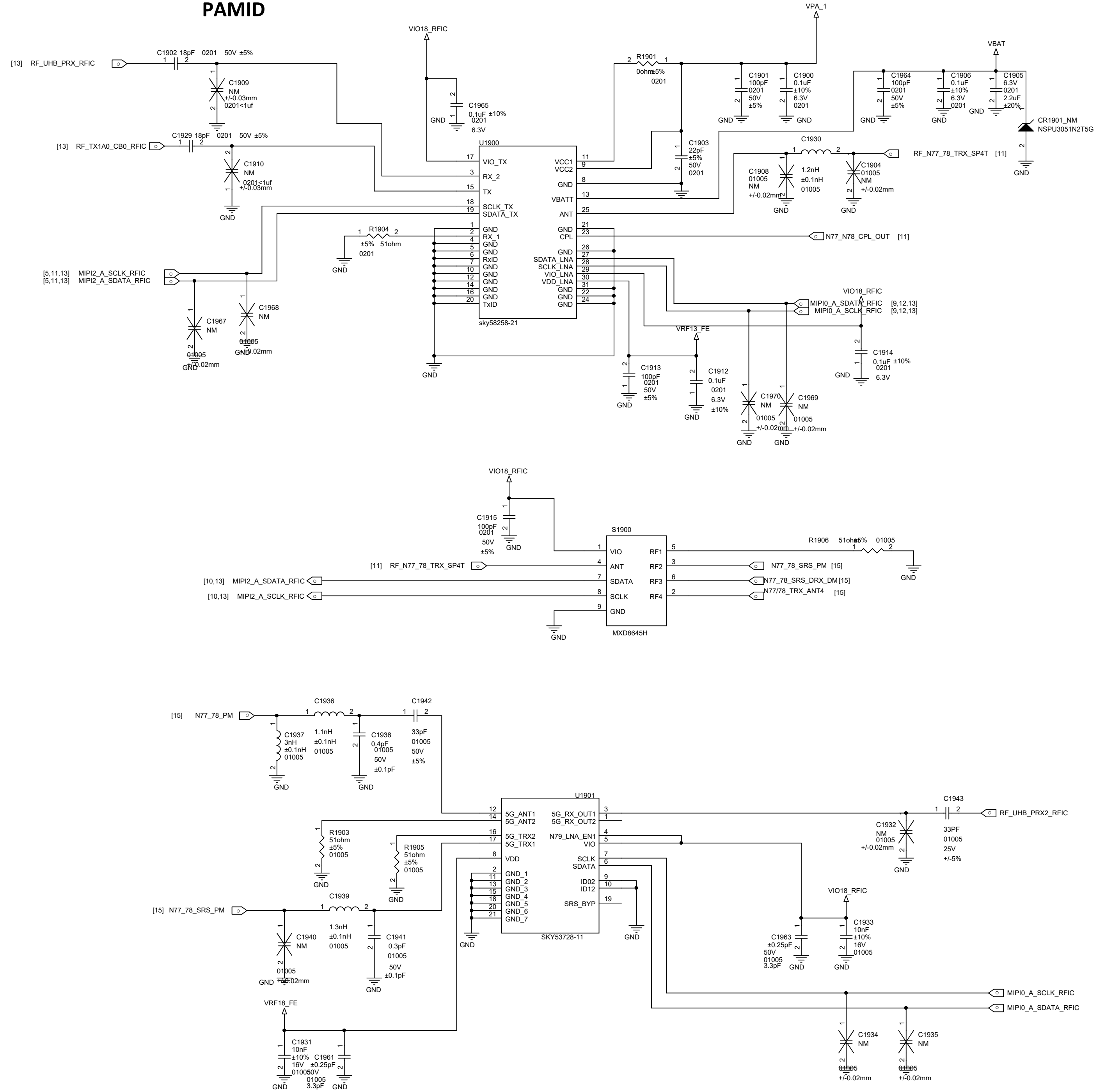
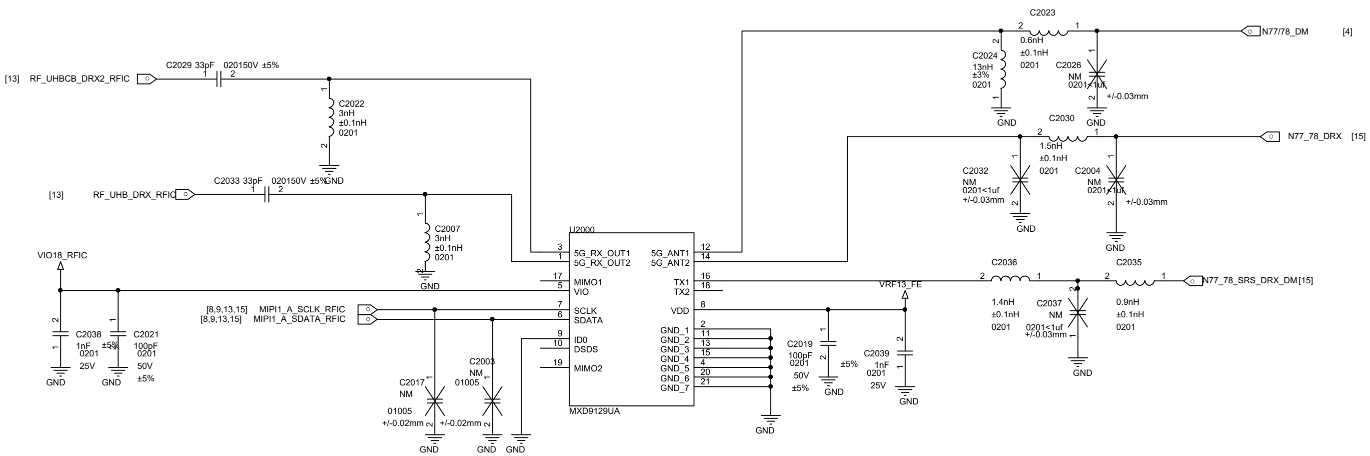
NR N77/N78 PAMID

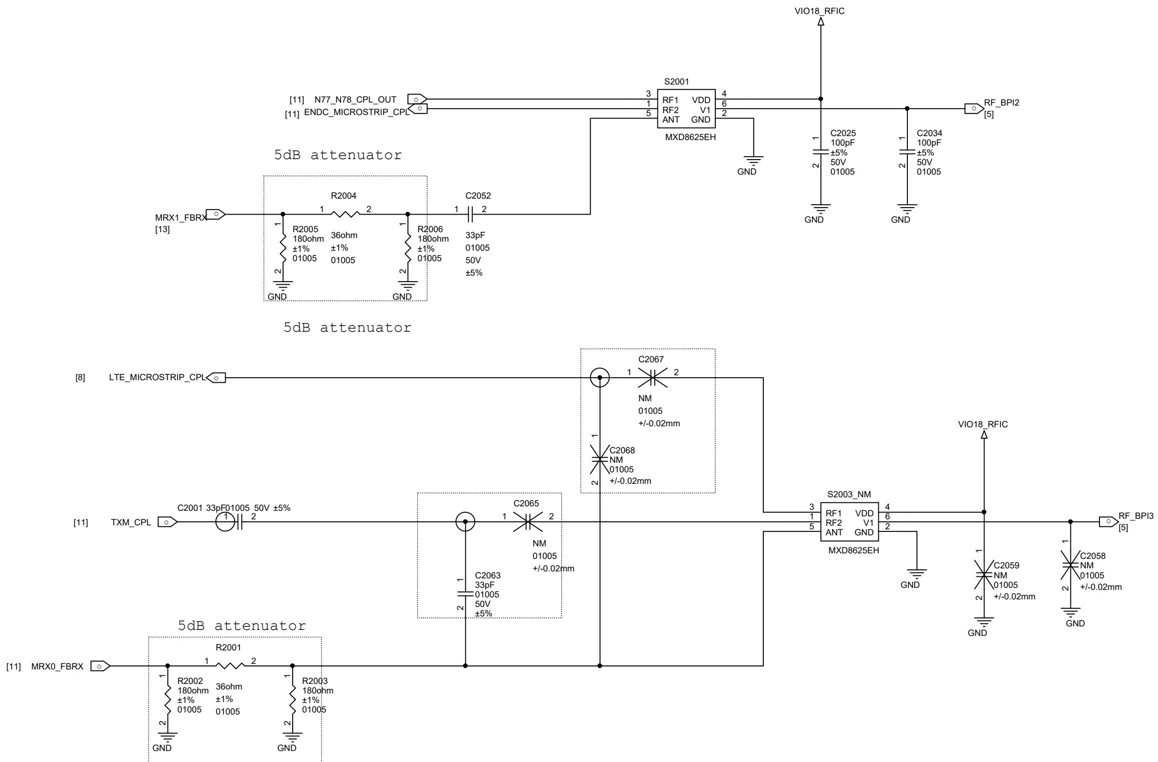


MI PHONE		
Title	K16_MAIN	
Size A2	Document Number Reserved(for 5G-WTR I)	Rev P0
Date: Tuesday, March 14, 2023	Sheet 19 of 76	

NR N77/N78/N79 DRx-LFEM



N1/3/7/40/41+N77/78 FBRX



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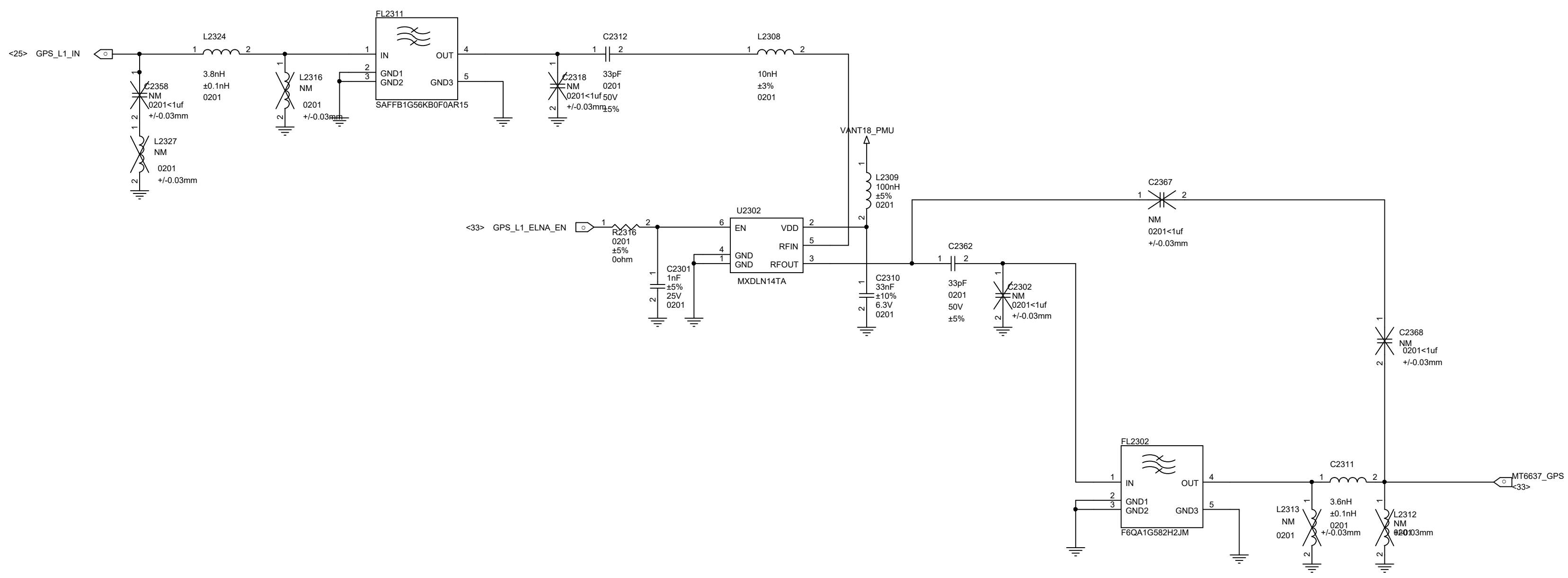
Size A2	Document Number Reserved(for 5G-WTR P)	Rev P0
Date: Tuesday, March 14, 2023	Sheet 20 of 76	



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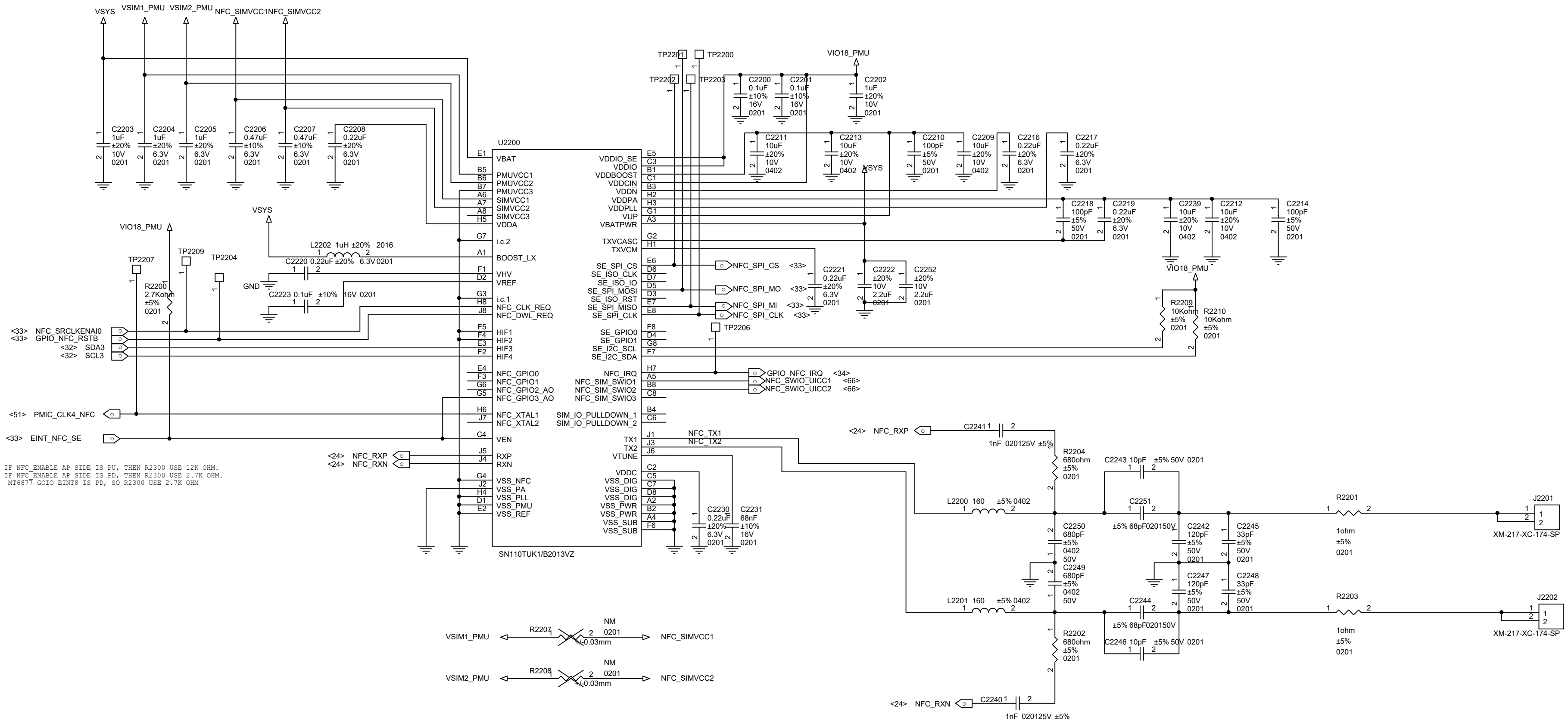
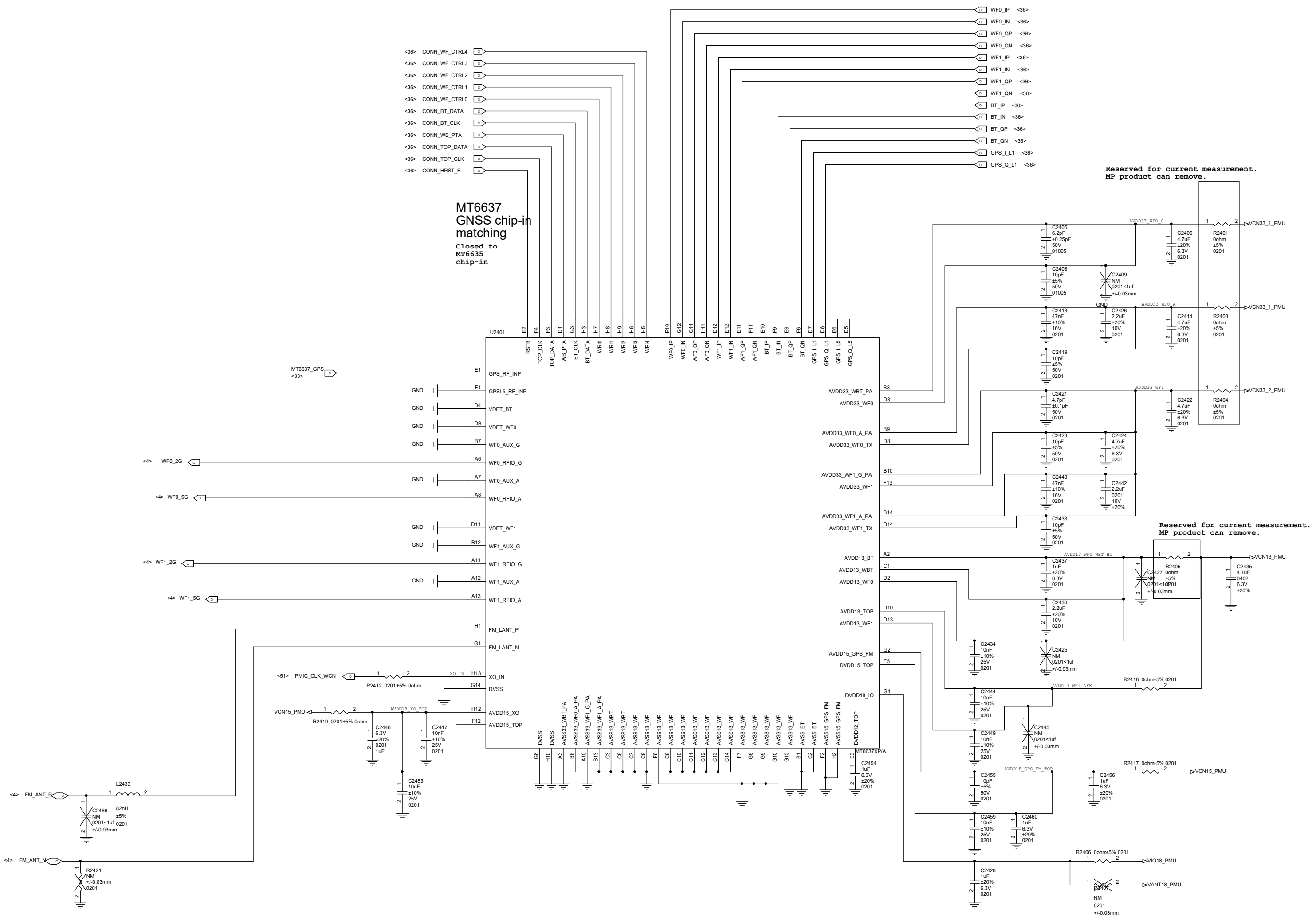
Title	MT6873_5GA	
Size A2	Document Number Reserved(for 5G-OTHERS)	Rev P0
Date: Tuesday, March 14, 2023	Sheet 23 of 74	



Title	K16_MAIN
Size A2	Document Number NFC	Rev P0
Date: Tuesday, March 14, 2023	Sheet 24 of 74



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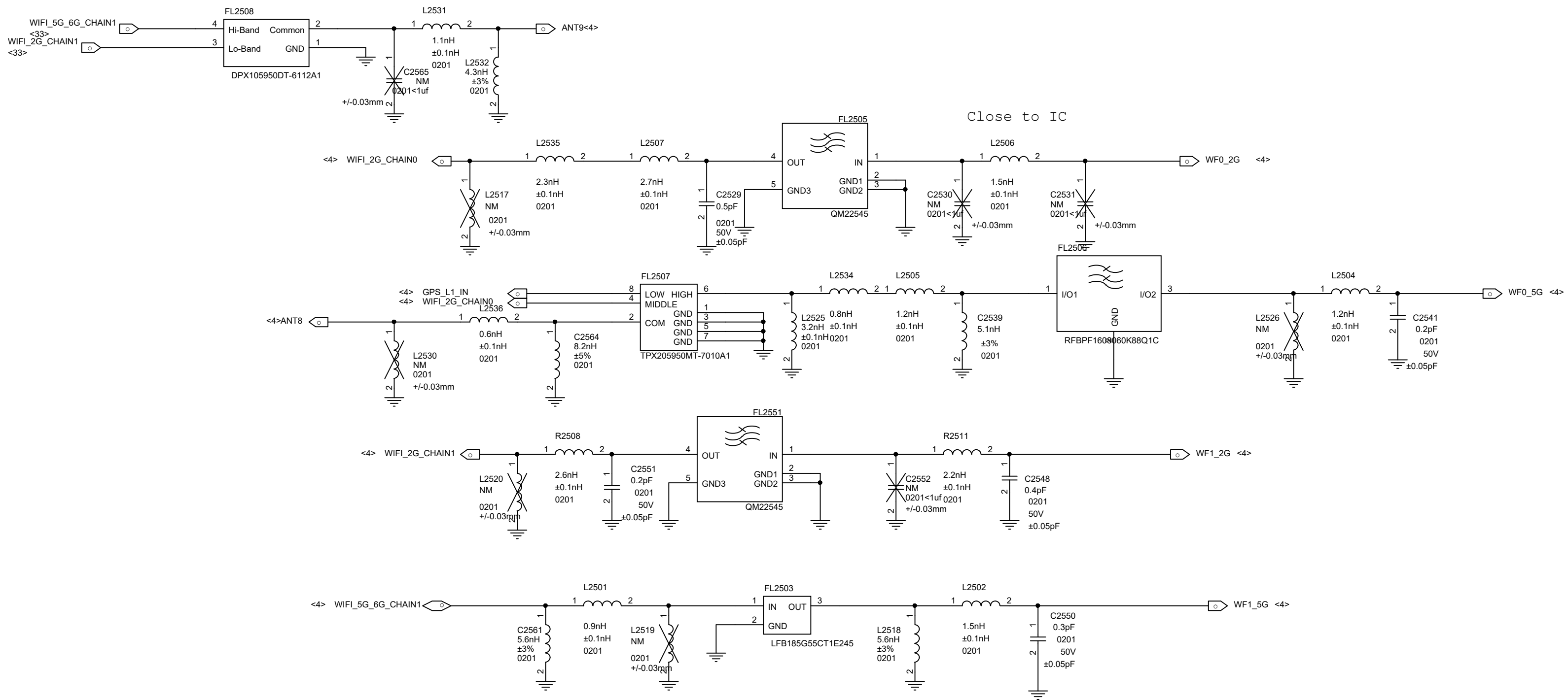
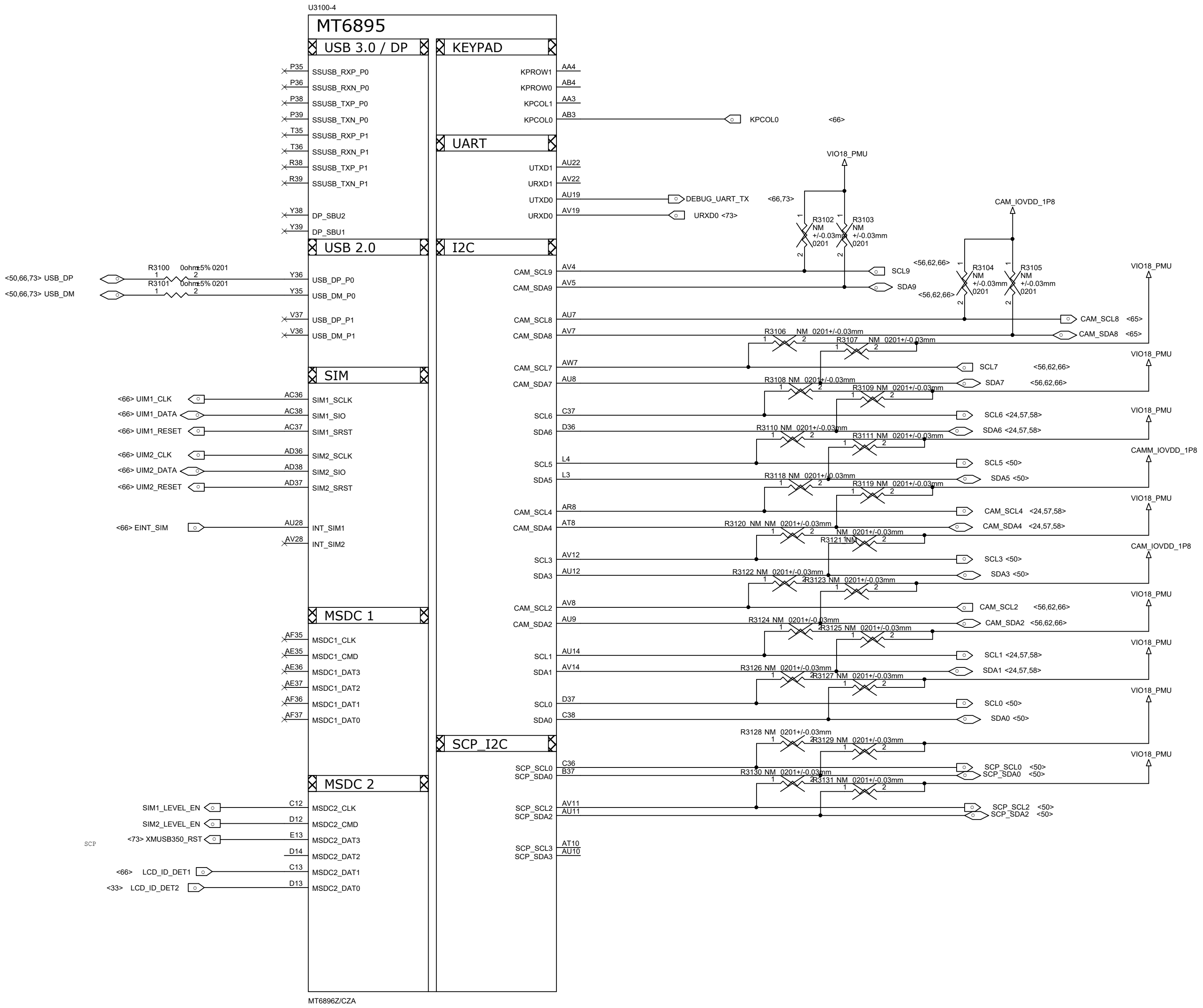
Title	MT6873_5GA	
Size A1	Document Number WIFI_Interface	Rev
Date: Tuesday, March 14, 2023	Sheet 25	of 74



MI PHONE		
Title		
Size A2	Document Number Reserved(for WIFI FEM)	Rev P0
Date: Tuesday, March 14, 2023	Sheet 26 of 74	



I2C9: SC8551A Master/CAMERA LDO

I2C8: WIDE_CAM

I2C7: FG/SC8551A Slave

I2C6: SMARTPA/FLASH/LCD

I2C5: MT6375/MT6691

I2C4:M_CAM

I2C3: NFC

I2C2: UW_CAM/F_CAM

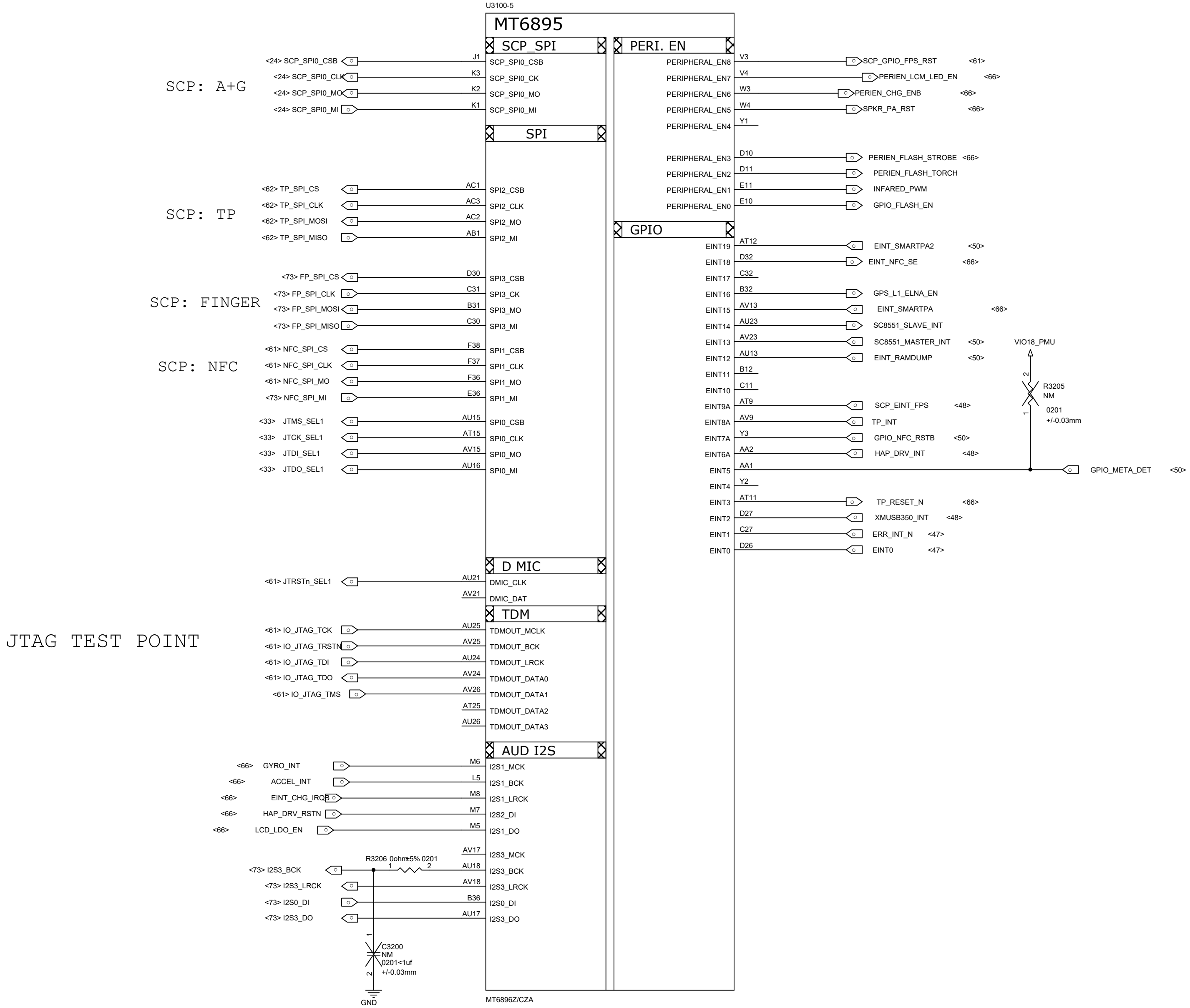
I2C1:HAPTIC

I2C0: XMUSB350

SCP_I2C0: L-SENSOR（前置）

SCP_I2C2: SAR/COMPASS/

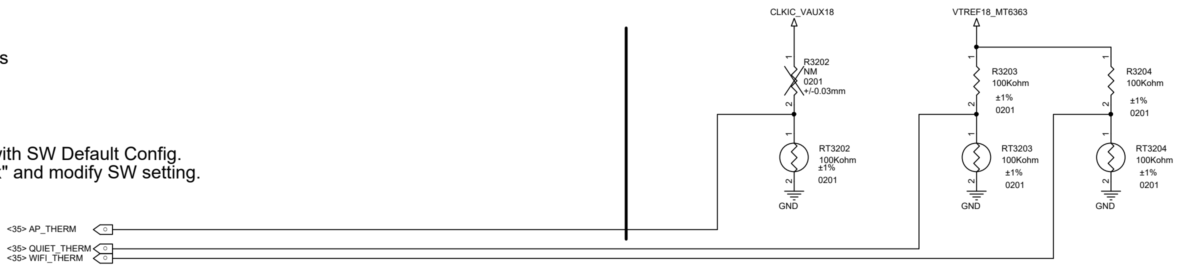
Title	MT6893_GPIO_1
Size A2	Document Number MT6893_GPIO_1
Date: Tuesday, March 14, 2023	Sheet 31 73



Schematic design notice:

Note 16-1: SW Default Configuration is as followings
AUX_IN2_NTC is for AP Temp.
AUX_IN3_NTC is for MHB PAMiD Temp.
AUX_IN4_NTC is for NR PAMiD Temp.

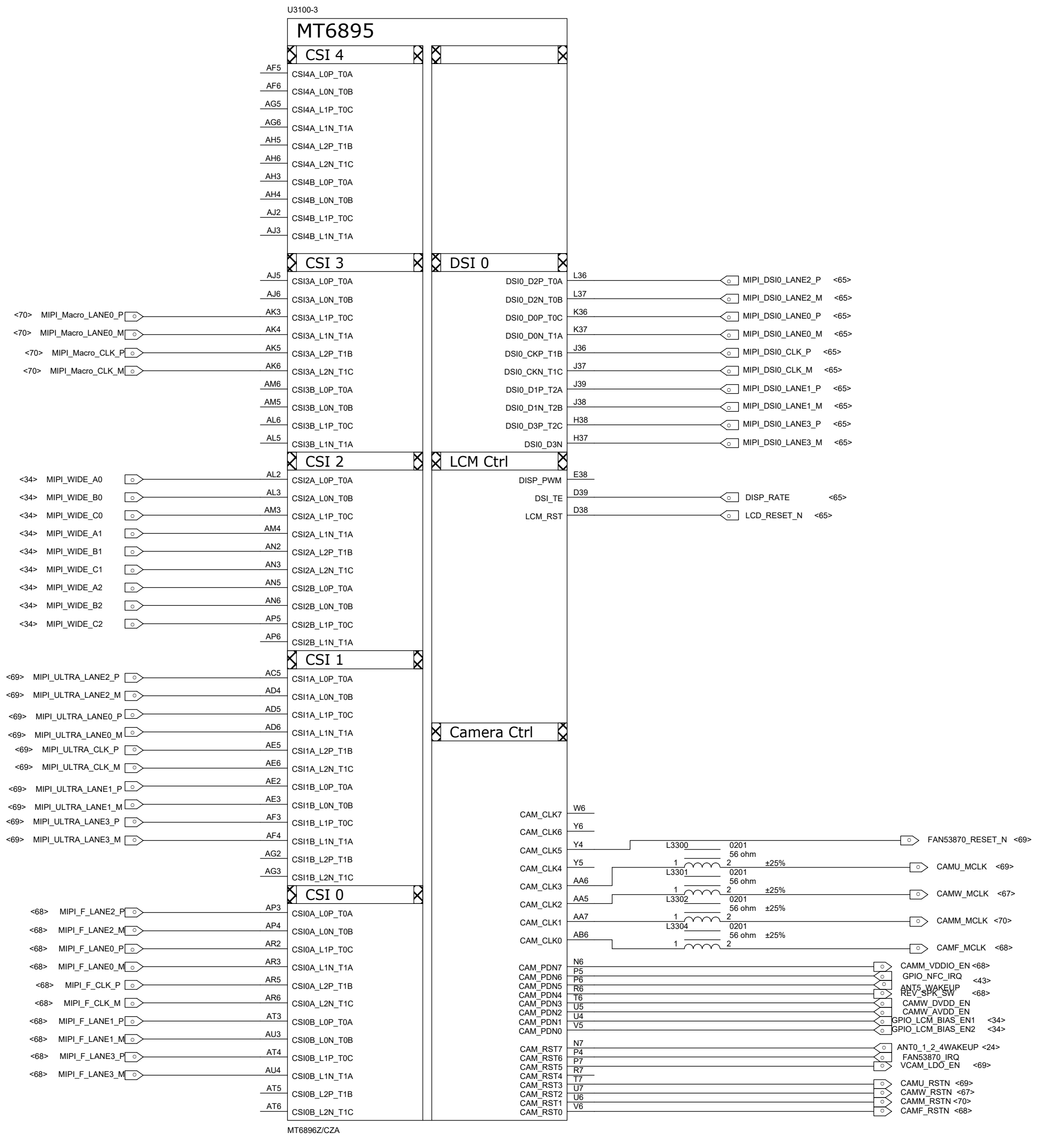
Note 16-2: If your design or placement is different with SW Default Config.
Please refer to "MTxxxx Thermal User Manuel.docx" and modify SW setting.



Thermistor to sense AP temperature

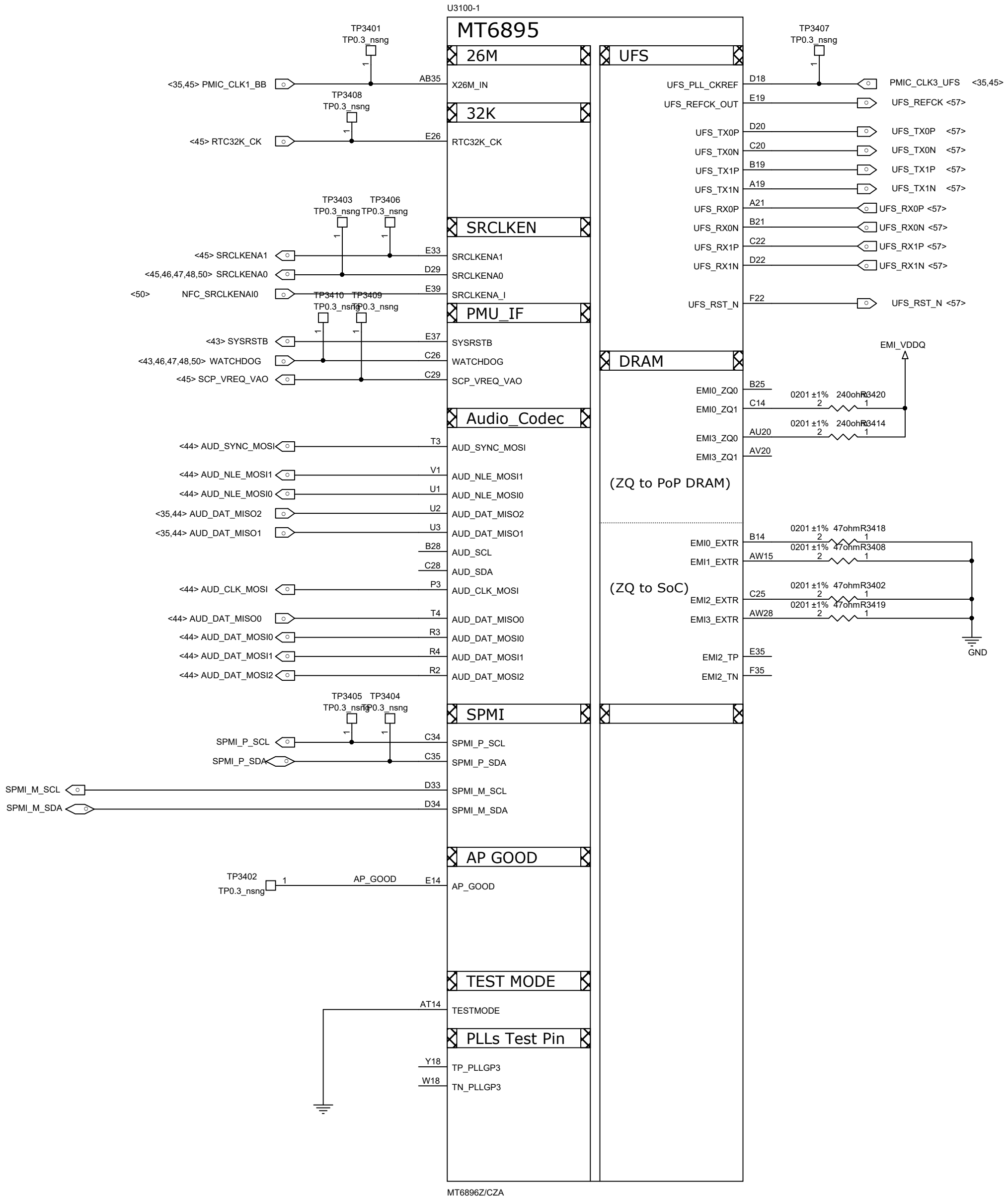
1. NTC1603 must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

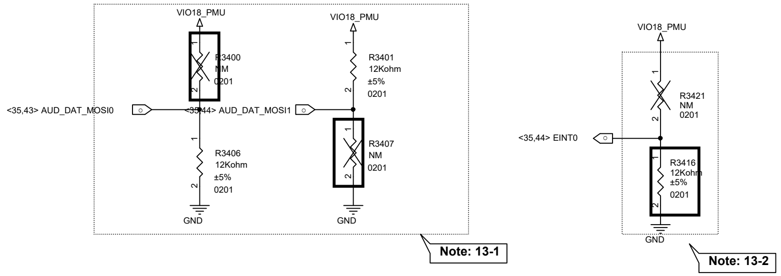
Title	MT6893_GPIO_2
Size A2	Document Number MT6893_GPIO_2
Date: Tuesday, March 14, 2023	Sheet 32 of 73



Title
MT6893_CAM_LCM

Size	Document Number	
C	MT6893_CAM_LCM	
Date: Tuesday, March 14, 2023		Sheet





Schematic design notice:

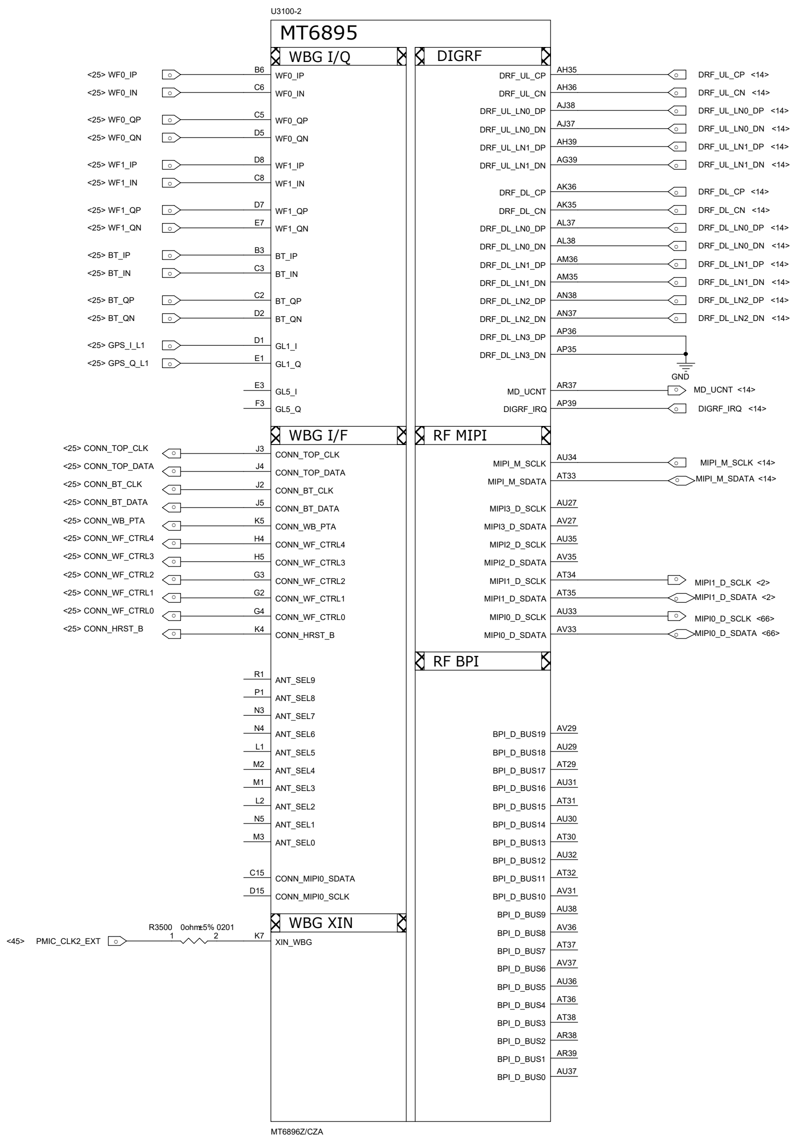
Note 13-1: "AUD_DAT_MOSI0" and "AUD_DAT_MOSI1"I pin features in trapping pin to enable JTAG.

AUD_DAT_MOSI0	AUD_DAT_MOSI1	AP JTAG	IO JTAG
L (Default)	L (Default)	N/A	N/A
L	H (by external PU)	SPI0_CSB, SPI0_CLK, SPI0_MO, SPI0_MI, DMIC_CLK	N/A
H (by external PU)	L	SPI0_CSB, SPI0_CLK, SPI0_MO, SPI0_MI, DMIC_CLK	TDM_LRCK/TDM_BCK/TDM_MCK/ TDM_DATA0/TDM_DATA1
H (by external PU)	H (by external PU)	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

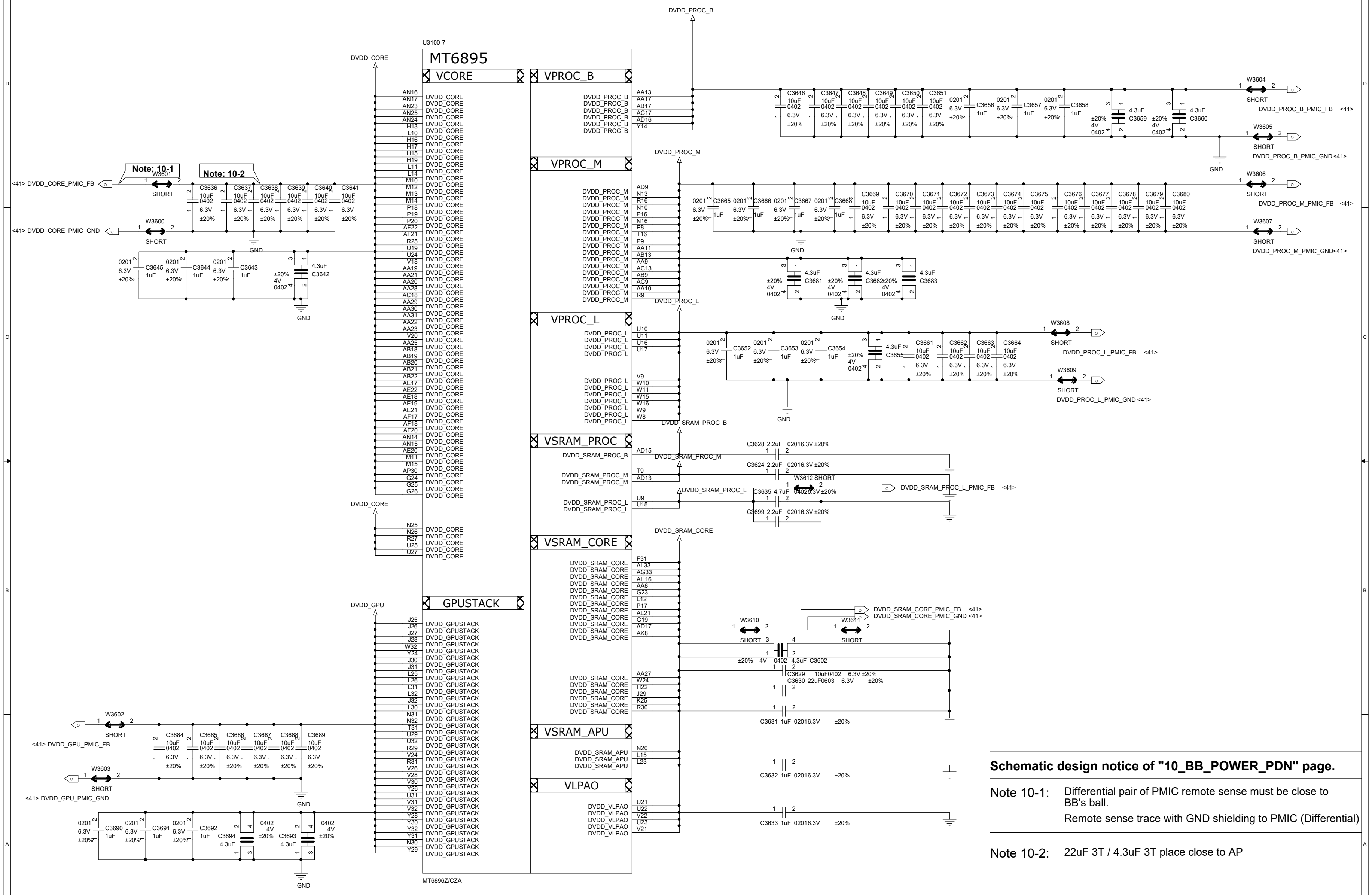
Note 13-2: "EINT0" ball is trapping pin to select DRAM Type for LP5-6400 and reserve device.

EINT0	DRAM Type
L (Default)	LP5-6400
H (by external PU)	Reserve

Title
MT6873_5GA
Size A2
Document Number
MT6893_INTF
Date: Tuesday, March 14, 2023
Sheet



Title	MT6893_RF_INTF
Size	C
Document Number	MT6893_RF_INTF
Date:	Tuesday, March 14, 2023
Sheet	35 of 73

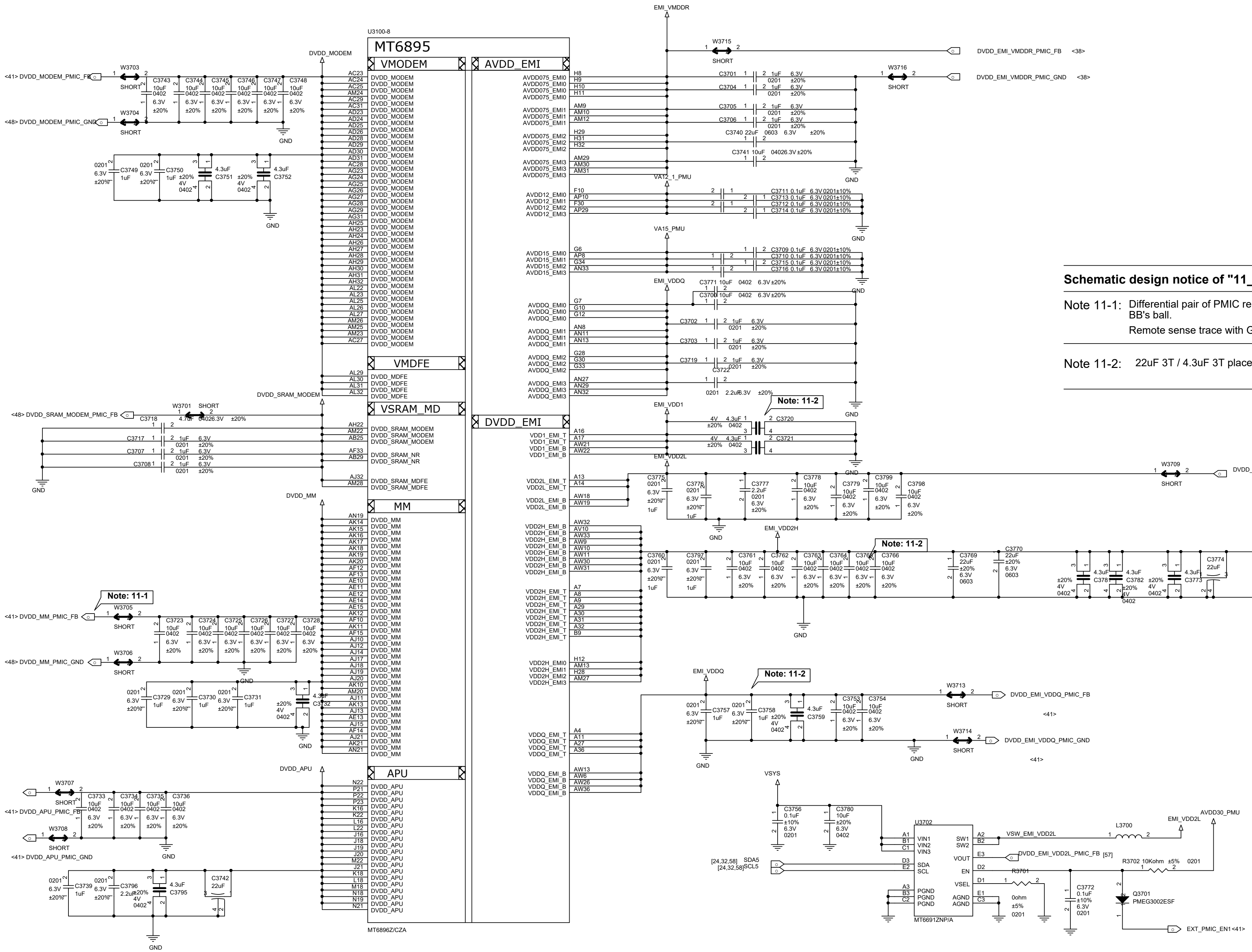


Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: 22uF 3T / 4.3uF 3T place close to AP

Title	MT6873_5GA
Size C	Document Number MT6893_PWR1
Date: Tuesday, March 14, 2023	Sheet 36 of 73

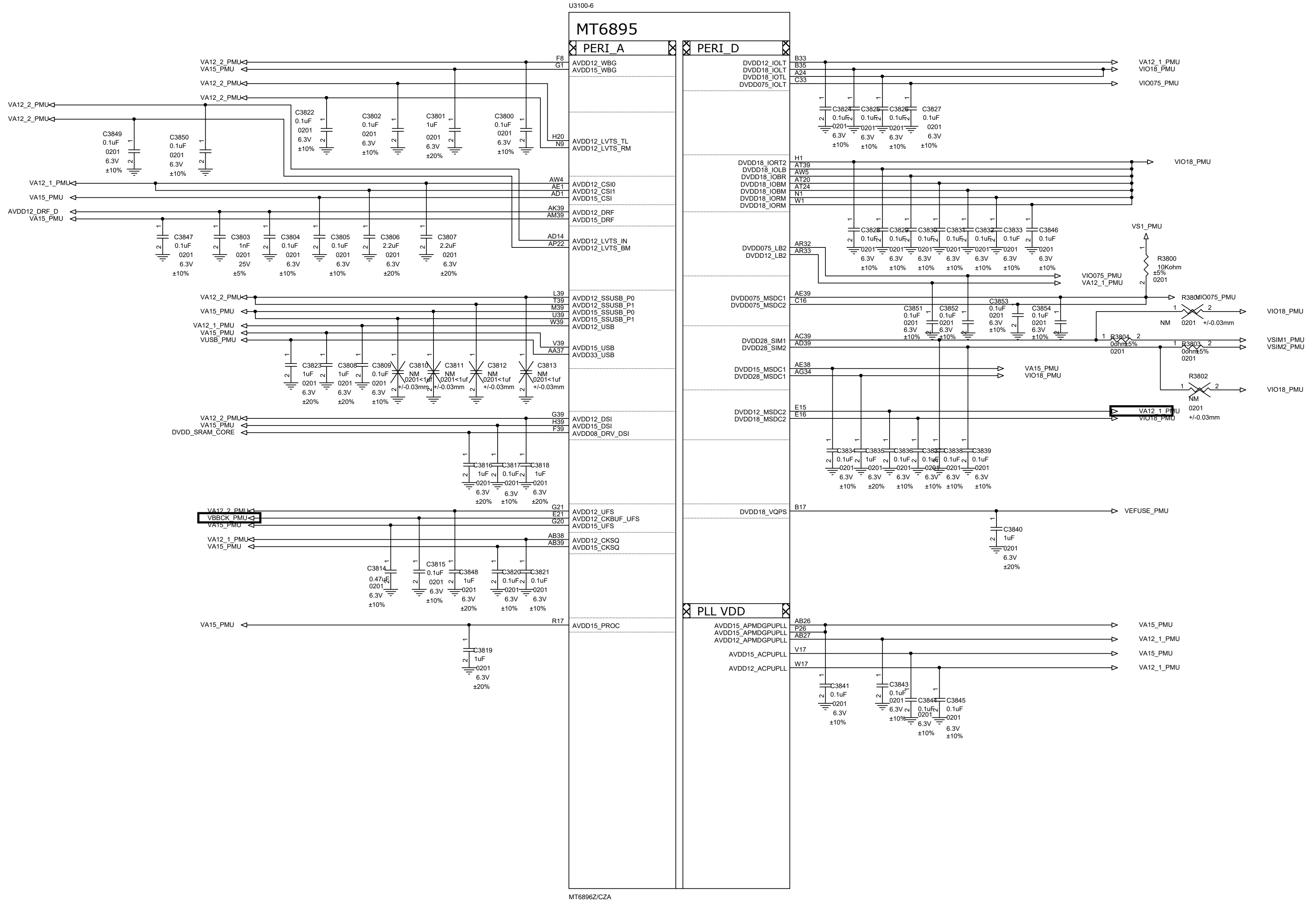


Schematic design notice of "11_BB_POWER_PDN" page.

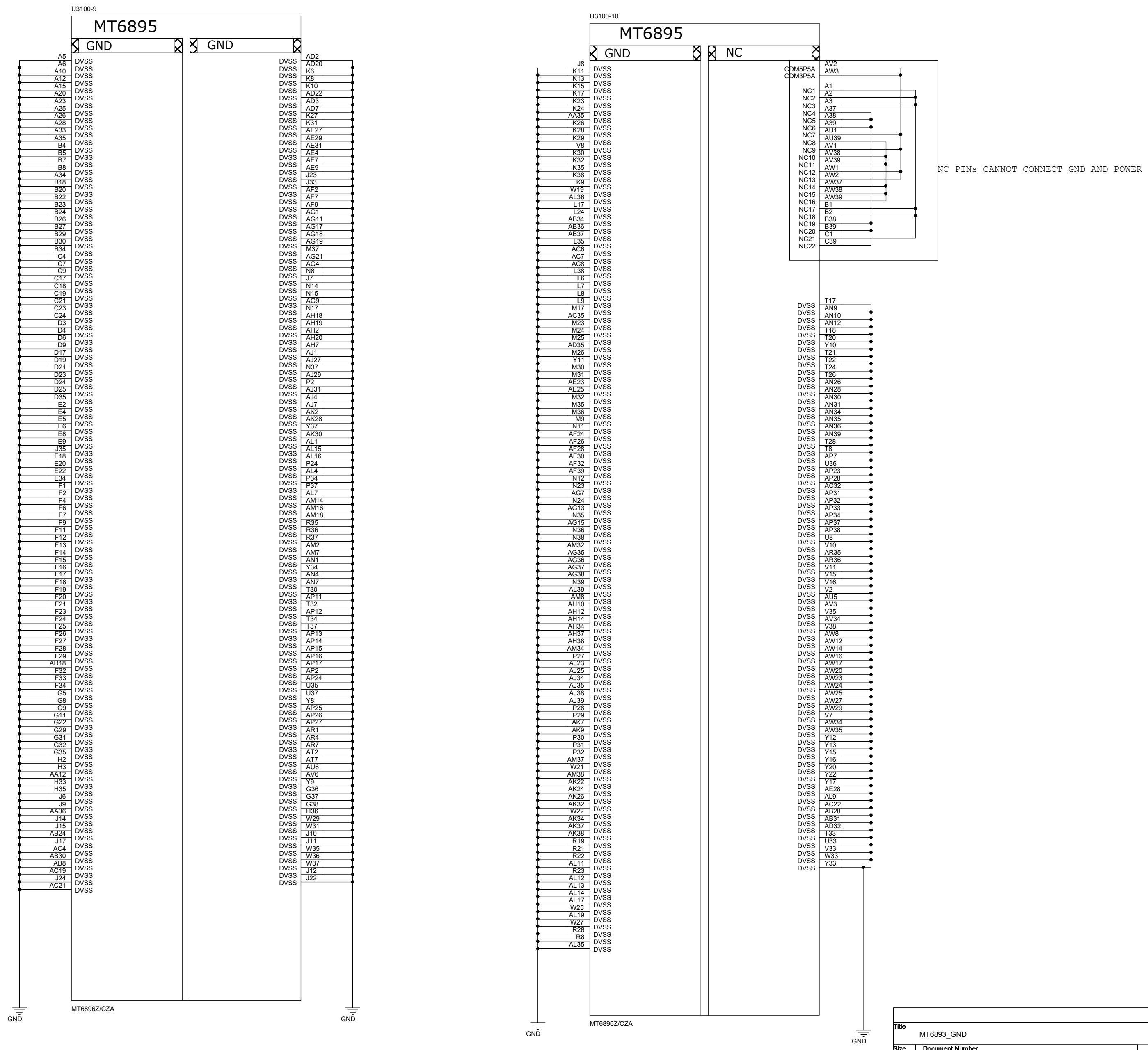
Note 11-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 11-2: 22uF 3T / 4.3uF 3T place close to AP

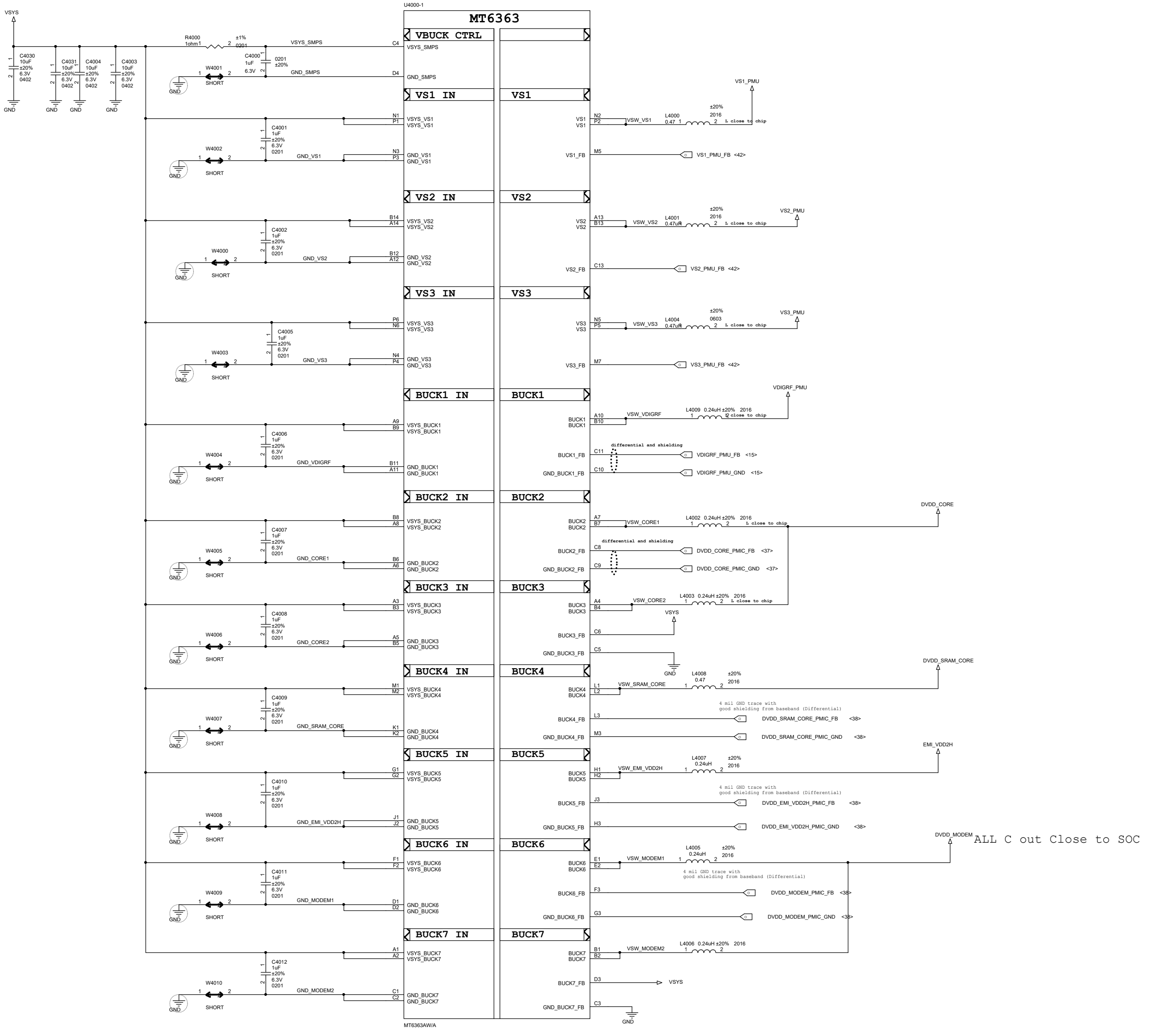
Title	MT6893_PWR2
Size A2	Document Number MT6893_PWR2
Date:	Tuesday, March 14, 2023
Sheet	



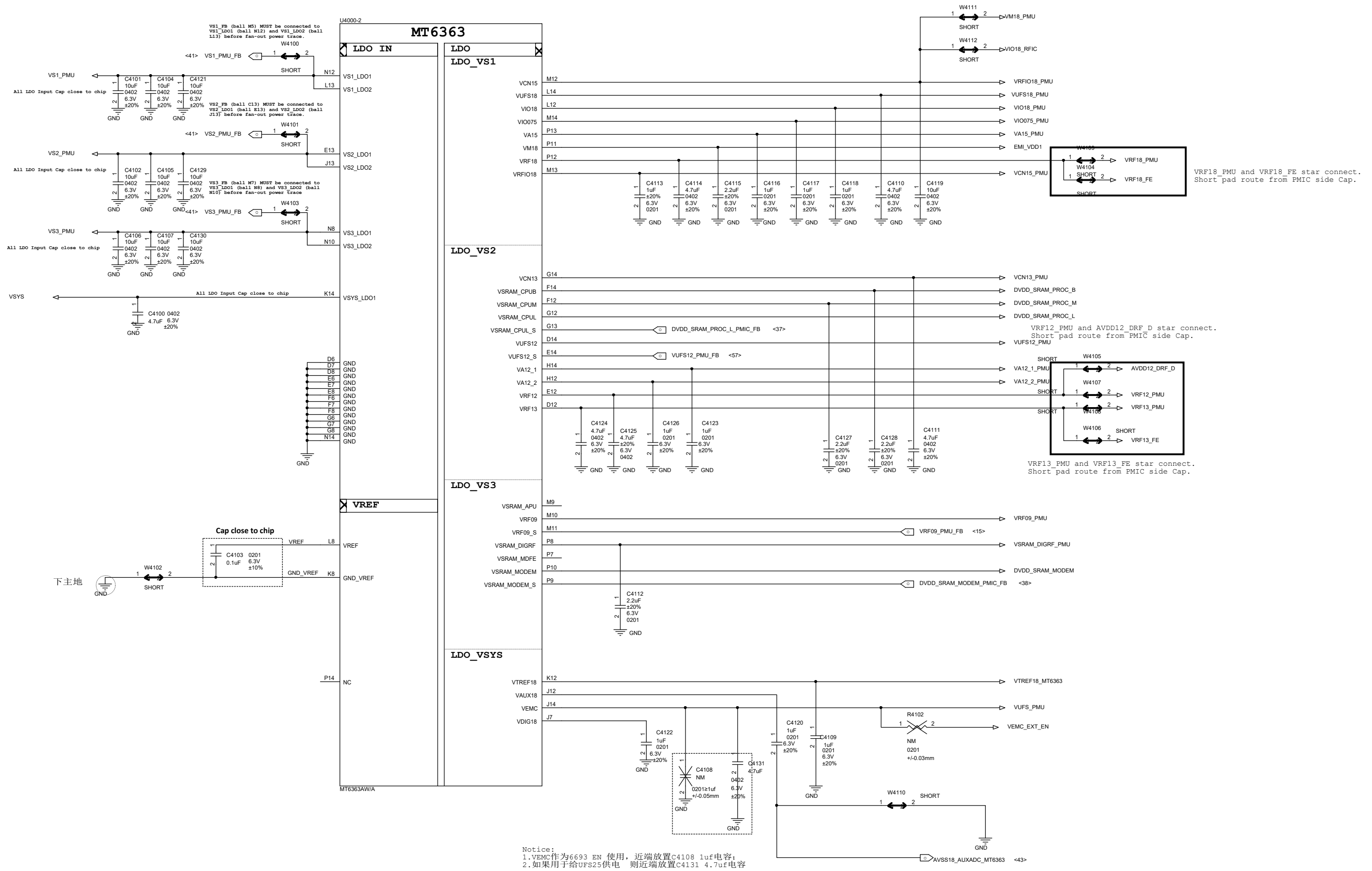
Title	MT6893_PWR3
Size A2	Document Number MT6893_PWR3
Date:	Tuesday, March 14, 2023
Sheet	38 of 73



Title	MT6893_GND
Size C	Document Number MT6893_GND
Date: Tuesday, March 14, 2023	Sheet 39 73



Title	MT6359_Buck
Size	A1
Document Number	MT6359_Buck
Date:	Tuesday, March 14, 2023
Sheet	40 of 73



Title	
Size	A1
Document Number	MT6359_LDO
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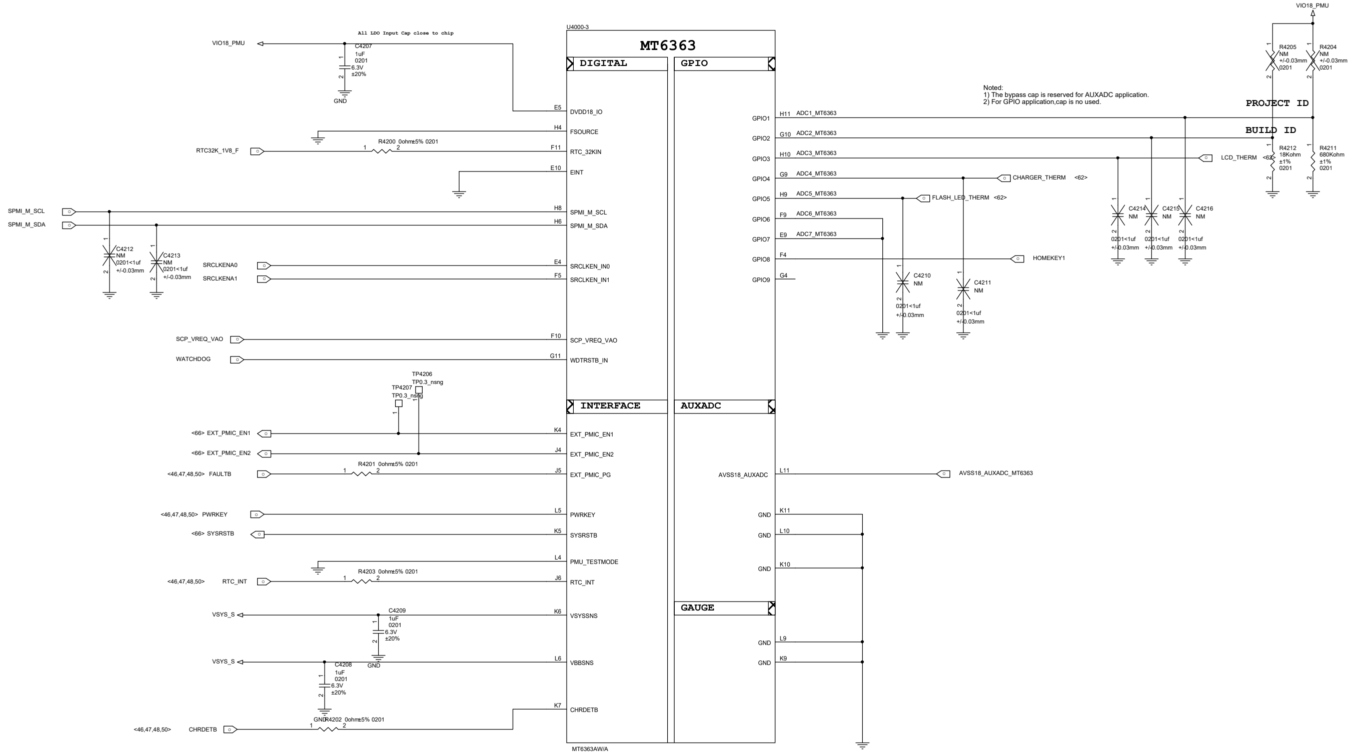
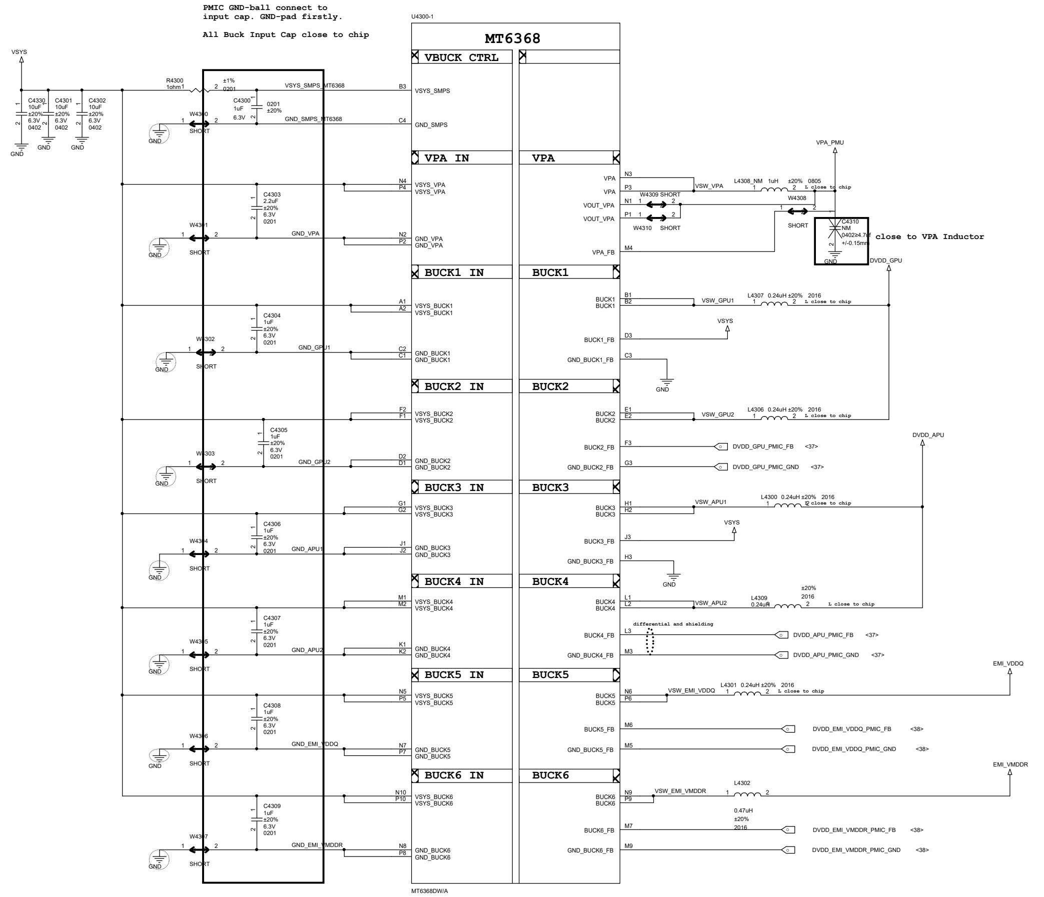
MT6363	Pin : EINT	VEMC
UFS3.1	0	2.55V
EMMC/UFS2.2	1	3V

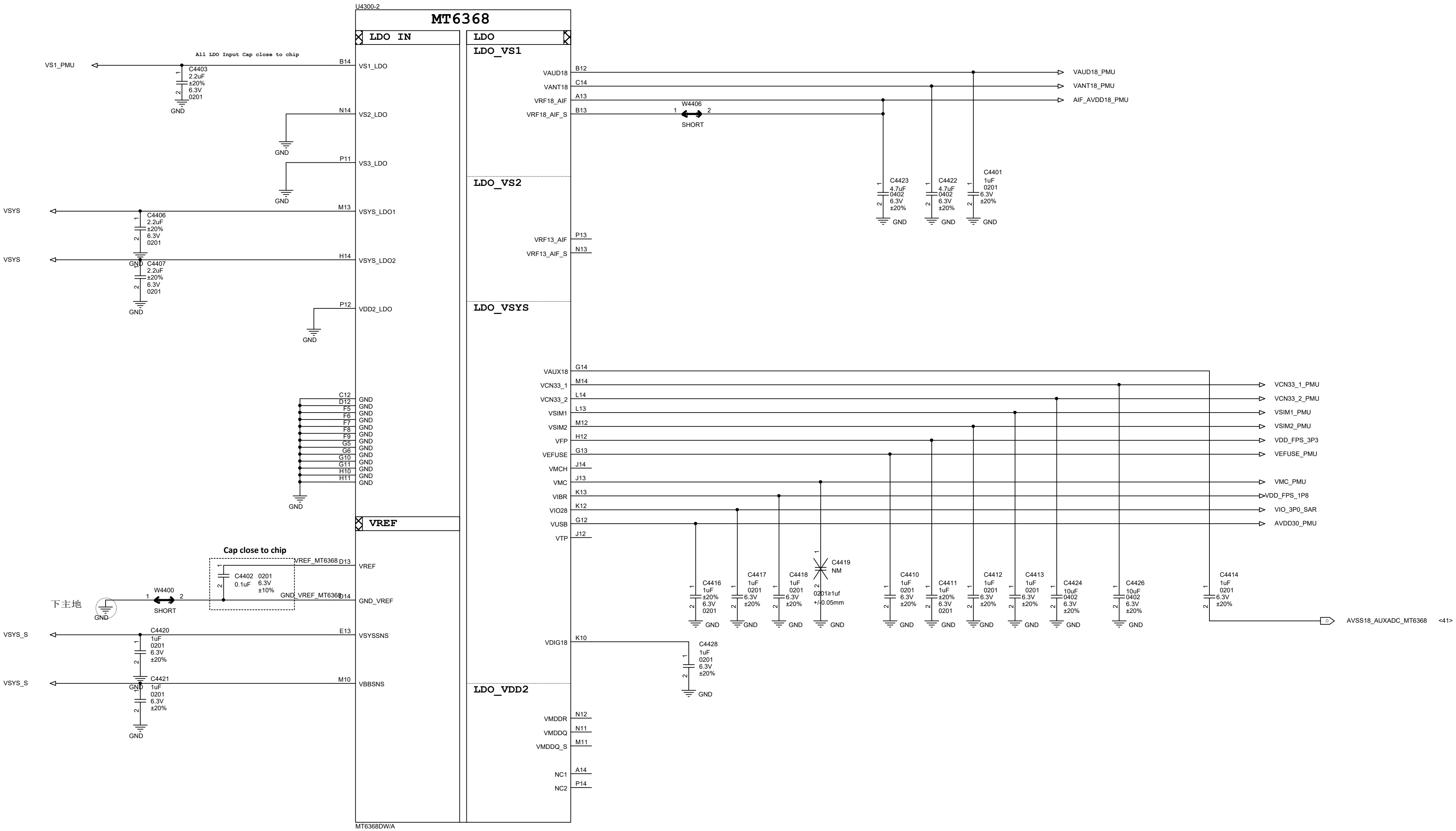
General MT6363
VEMC selection HW TRAPPING



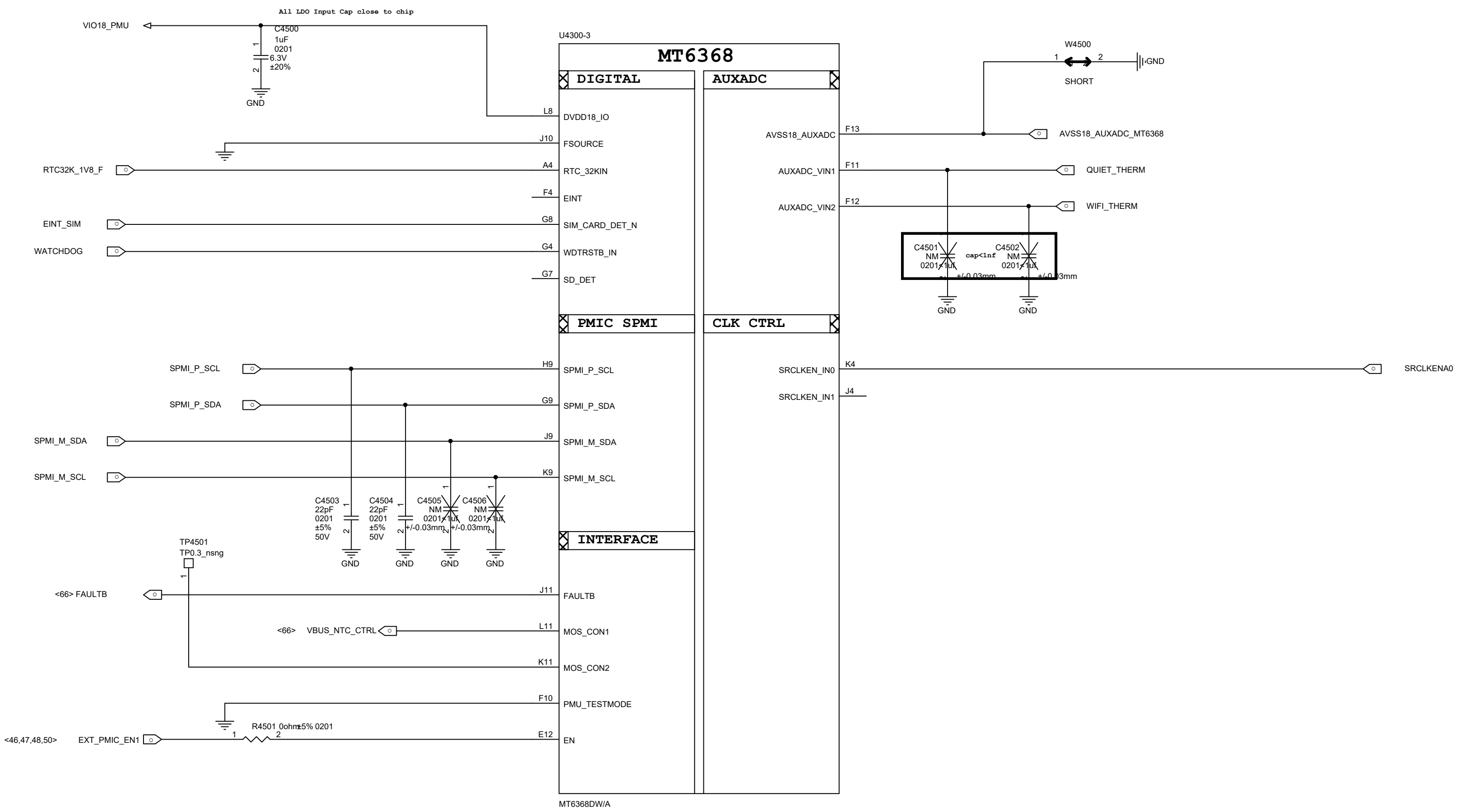
Title	MT6359_General
Size A1	Document Number MT6359_General
Date: Tuesday, March 14, 2023	Sheet 42 of 73



Title	MT6359_AUDIO
Size A1	Document Number MT6359_AUDIO
Date: Tuesday, March 14, 2023	Sheet 43 of 73



Title	MT6359_Clock
Size A2	Document Number MT6359_Clock
Date: Tuesday, March 14, 2023	Sheet 44 73

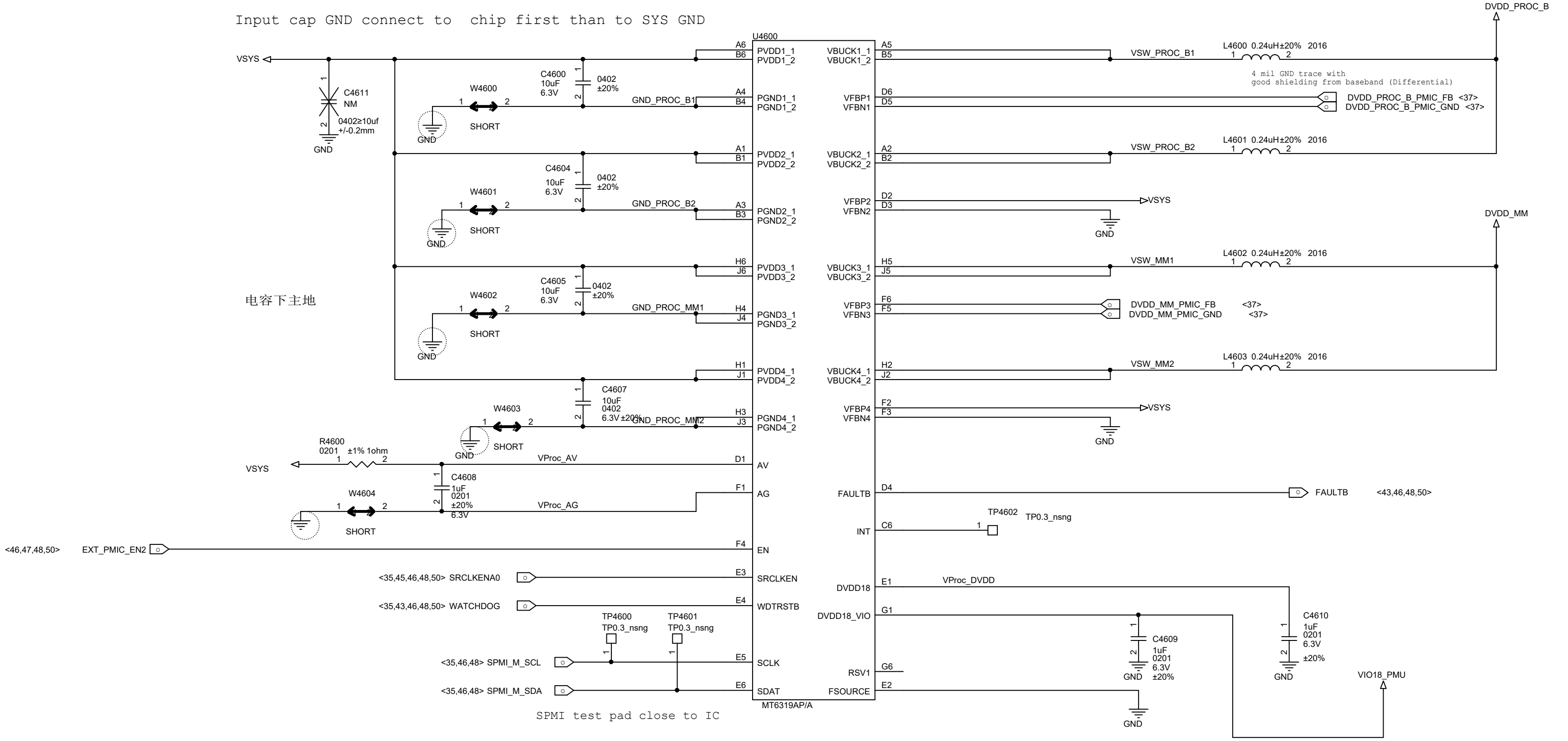


Title	MT6315_VGPU
Size A2	Document Number MT6315_VGPU
Date: Tuesday, March 14, 2023	Sheet 45 73

MT6319 4-Phase Buck (CPU)

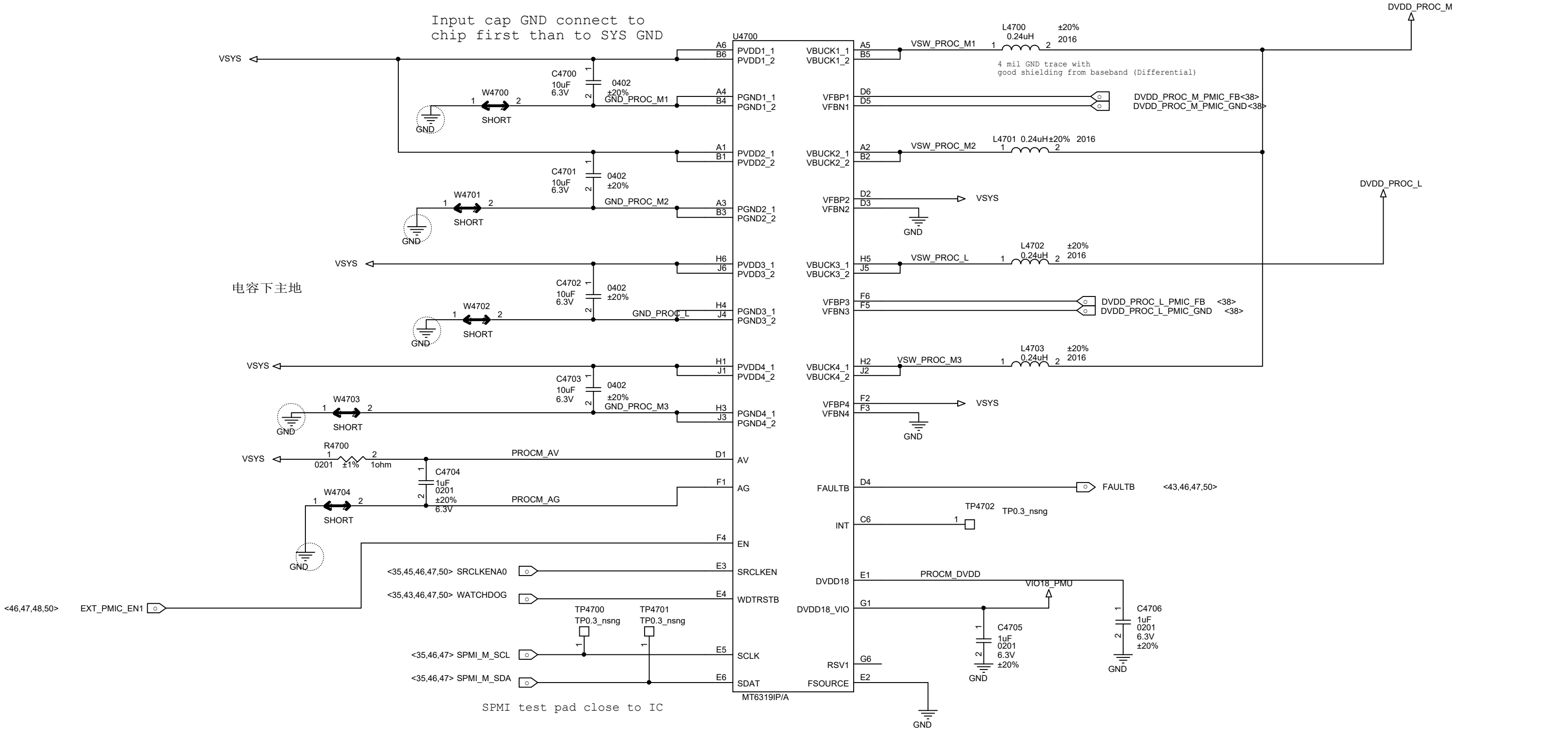
ALL C out Close to SOC

Input cap GND connect to chip first than to SYS GND

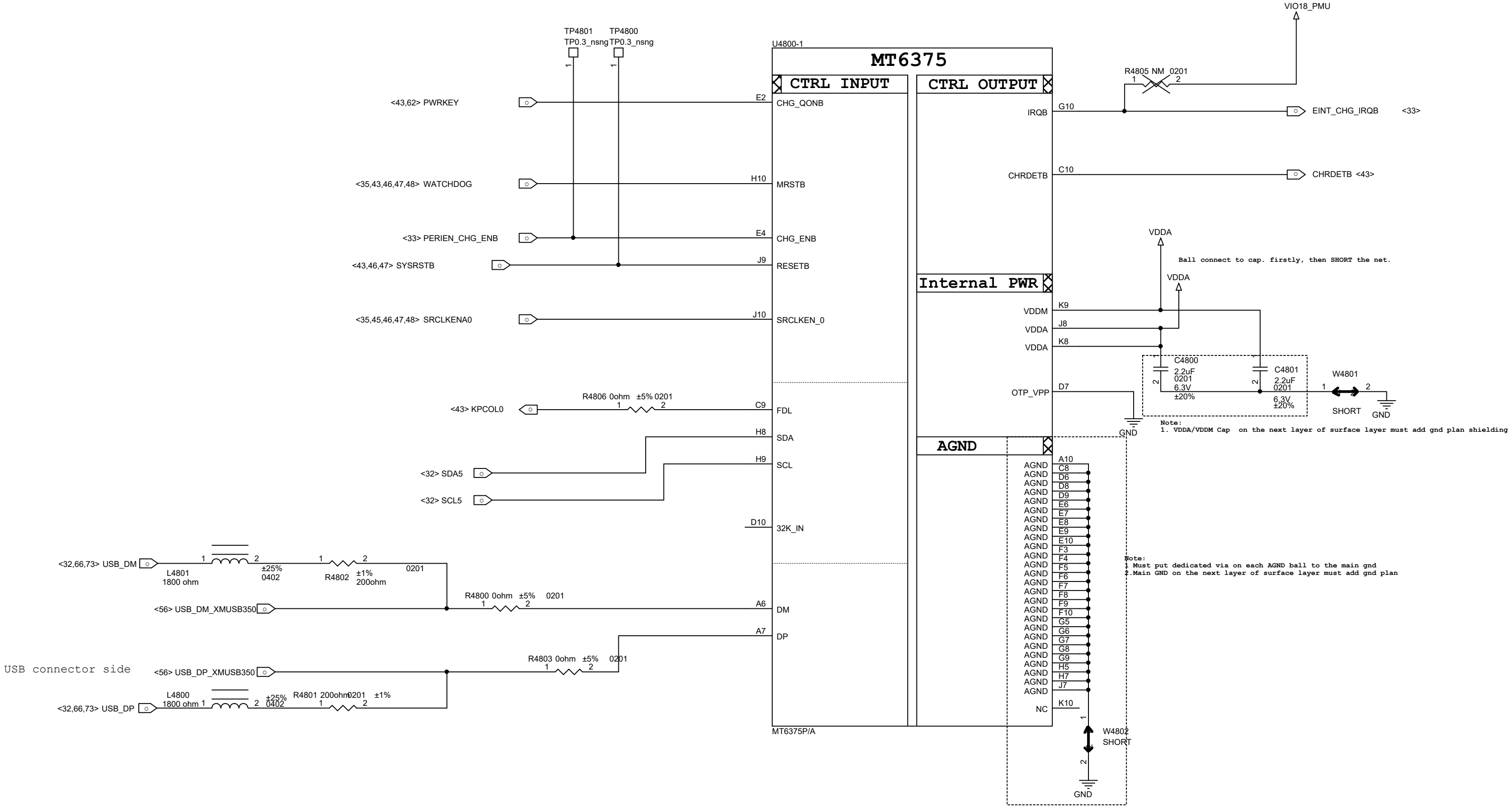


Title	MT6315_VCPU	
Size A2	Document Number MT6315_VCPU	
Date: Tuesday, March 14, 2023	Sheet 46	73

MT6319 4-Phase Buck (PROC_M)



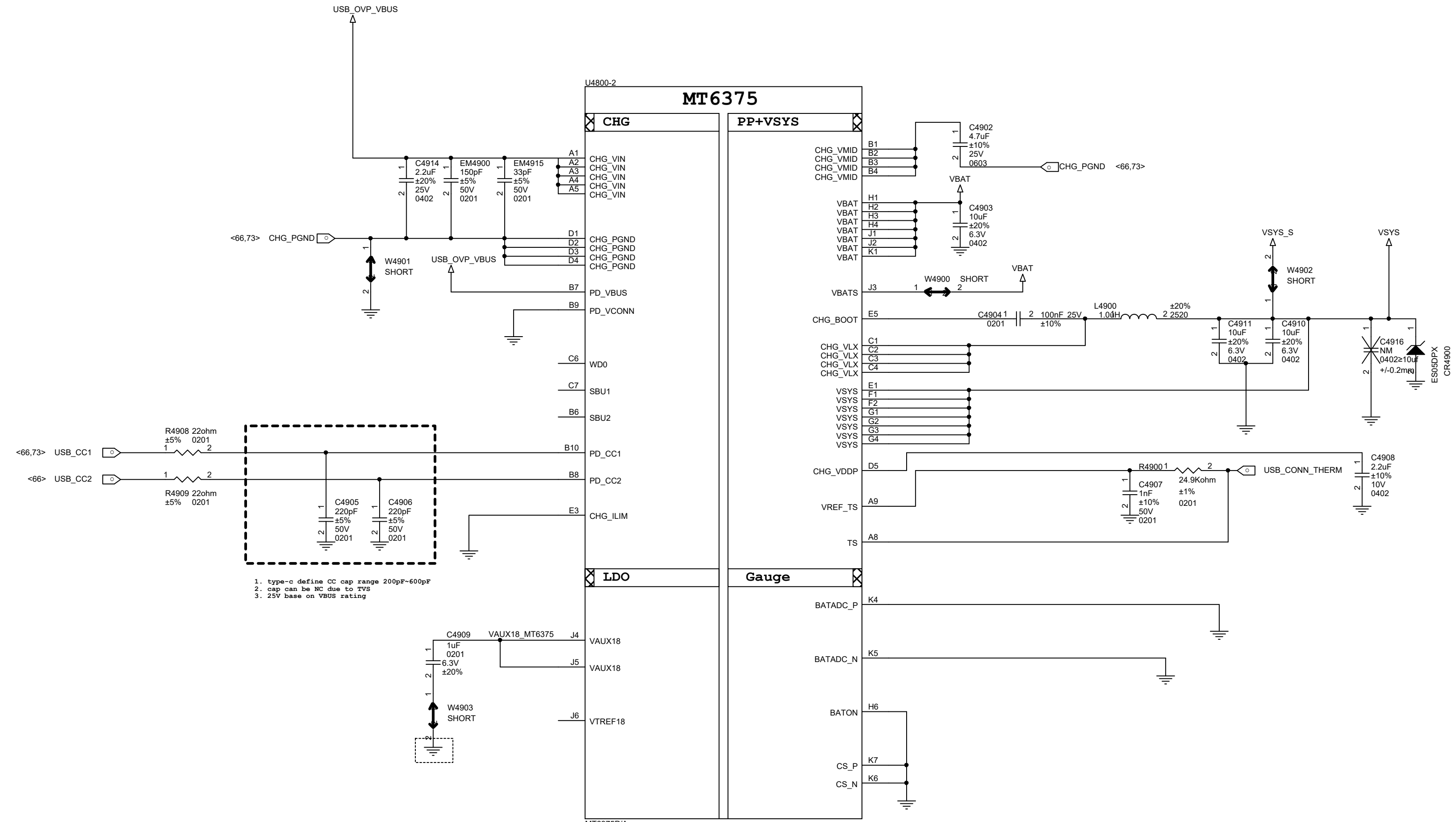
Title	MT6315_VMD	
Size A2	Document Number MT6315_VMD	
Date: Tuesday, March 14, 2023	Sheet 47	of 73



Title MT6360_BUCK_LDO

Size A2 | Document Number MT6360_BUCK_LDO

Date: Tuesday, March 14, 2023 | Sheet



Note: VAUX18/VTREF18 Cap on the next layer of surface layer must add gnd plan shielding

Title				
Size A2	Document Number MT6360_General			
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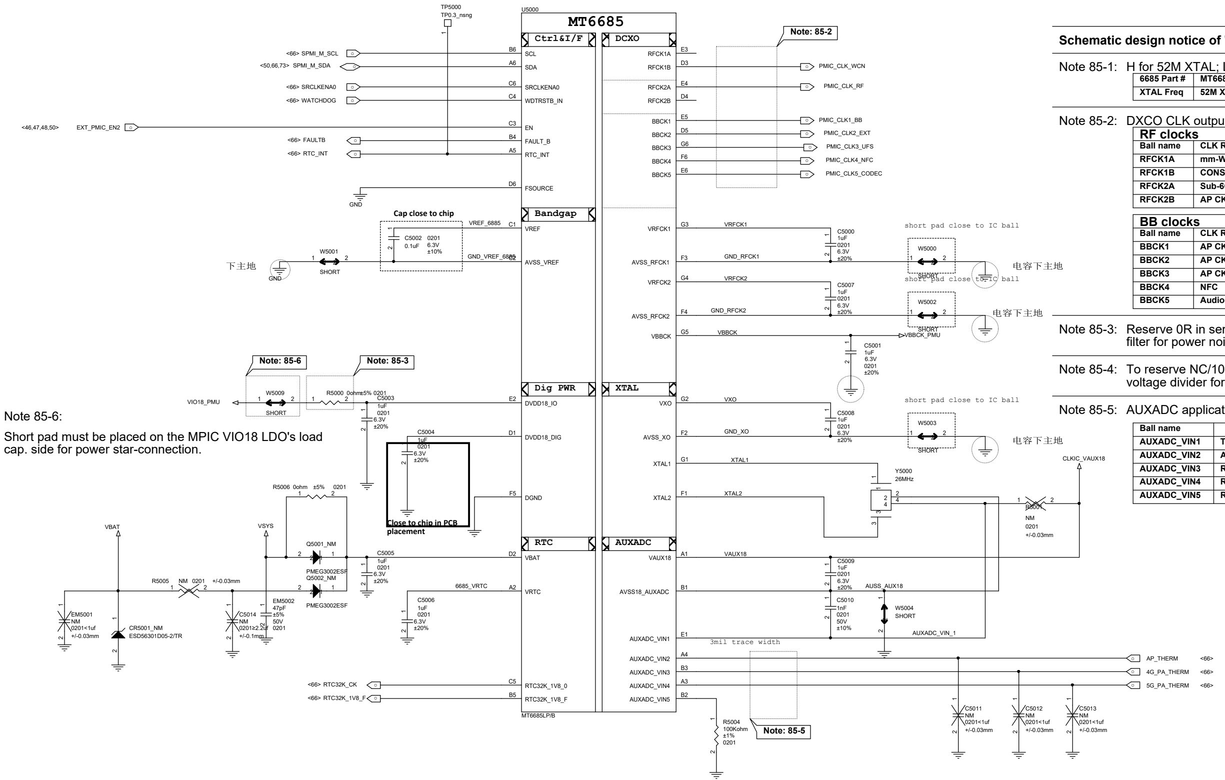

Note 85-6:

Short pad must be placed on the MPIC VIO18 LDO's load cap. side for power star-connection.

Schematic design notice of "81_MT6685 CLK AUXADC" page.

Note 85-1: H for 52M XTAL; L for H for 26M XTAL

6685 Part #	MT6685HP/A	MT6685LP/A
XTAL Freq	52M XTAL	26M XTAL

Note 85-2: DXCO CLK output assignment to CLK receiver

RF clocks

Ball name	CLK Receiver	Freq.(MHz)
RFCK1A	mm-Wave RF	XTAL (26/52)
RFCK1B	CONSYS RF	XTAL (26/52)
RFCK2A	Sub-6G RF	26M
RFCK2B	AP CK4 (mm-Wave ABB)	26M

BB clocks

Ball name	CLK Receiver	Freq.(MHz)
BBCK1	AP CK1 (System-boot)	26M
BBCK2	AP CK2 (WBG)	26M
BBCK3	AP CK3 (UFS)	26M
BBCK4	NFC	26M
BBCK5	Audio CODEC	26M

Note 85-3: Reserve 0R in series connection to form a R/C low pass filter for power noise reduction

Note 85-4: To reserve NC/100K ohm(1%) on AUXADC_VIN1 as voltage divider for temperature sensor in TMS

Note 85-5: AUXADC application

Ball name	Application
AUXADC_VIN1	TMS
AUXADC_VIN2	AP temperature measurement in board level
AUXADC_VIN3	RF PA1 temperature measurement
AUXADC_VIN4	RF PA2 temperature measurement
AUXADC_VIN5	Reserved. (Tied to GND for not used AUXADC)

Title	MT6360_Charger
Size A1 / Document Number	MT6360_Charger
Date:	Tuesday, March 14, 2023
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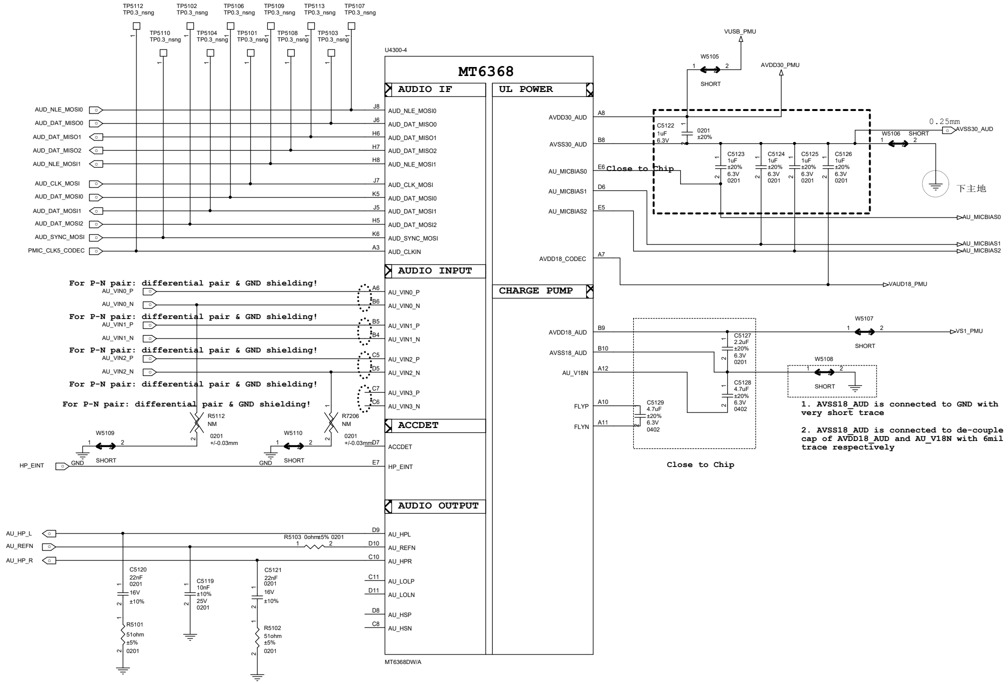

-AU_HPL and AU_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively

-The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"

Title	MT6338_CODEC
Size	A1
Document Number	MT6338_CODEC
Date:	Tuesday, March 14, 2023
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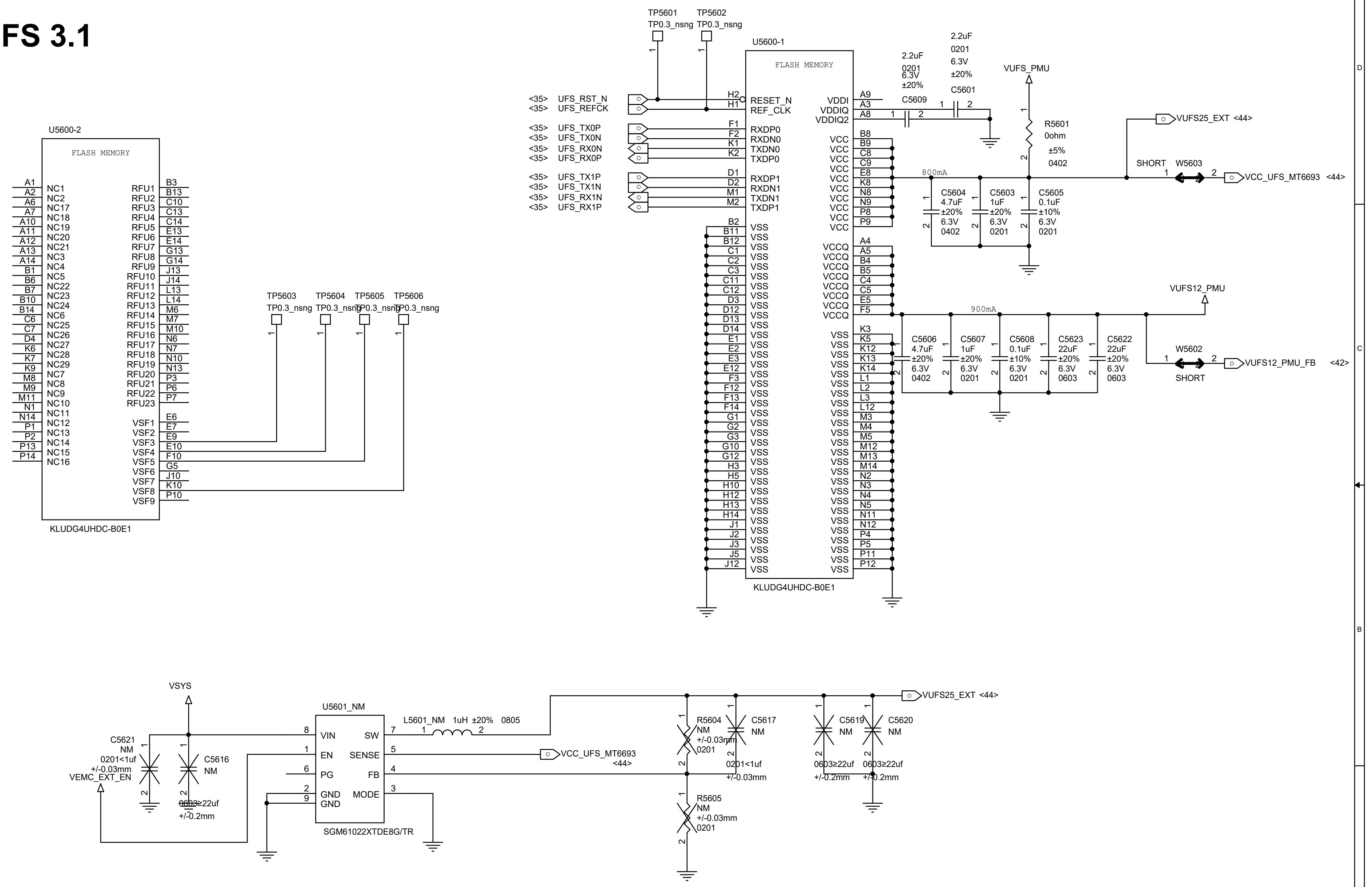
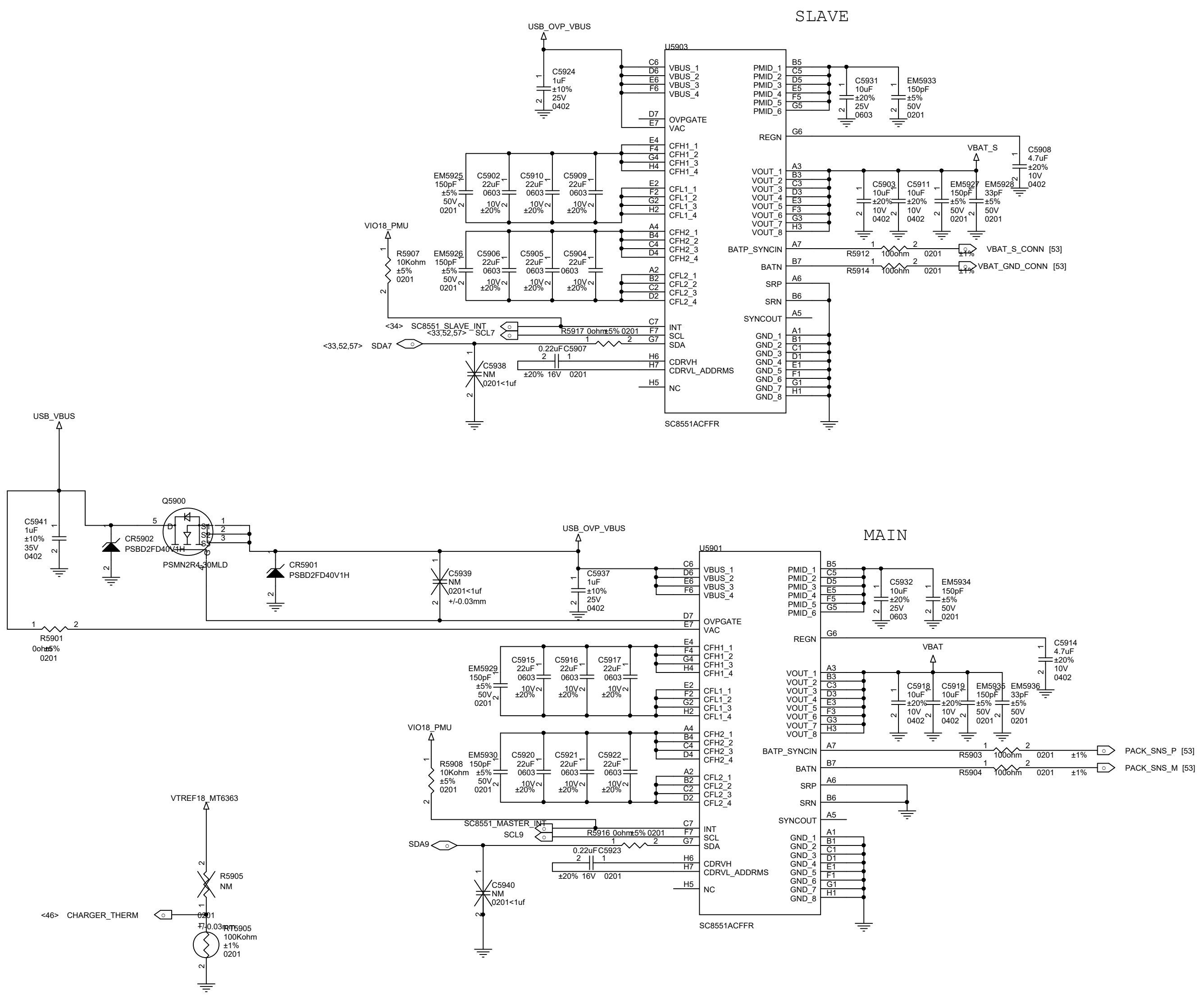
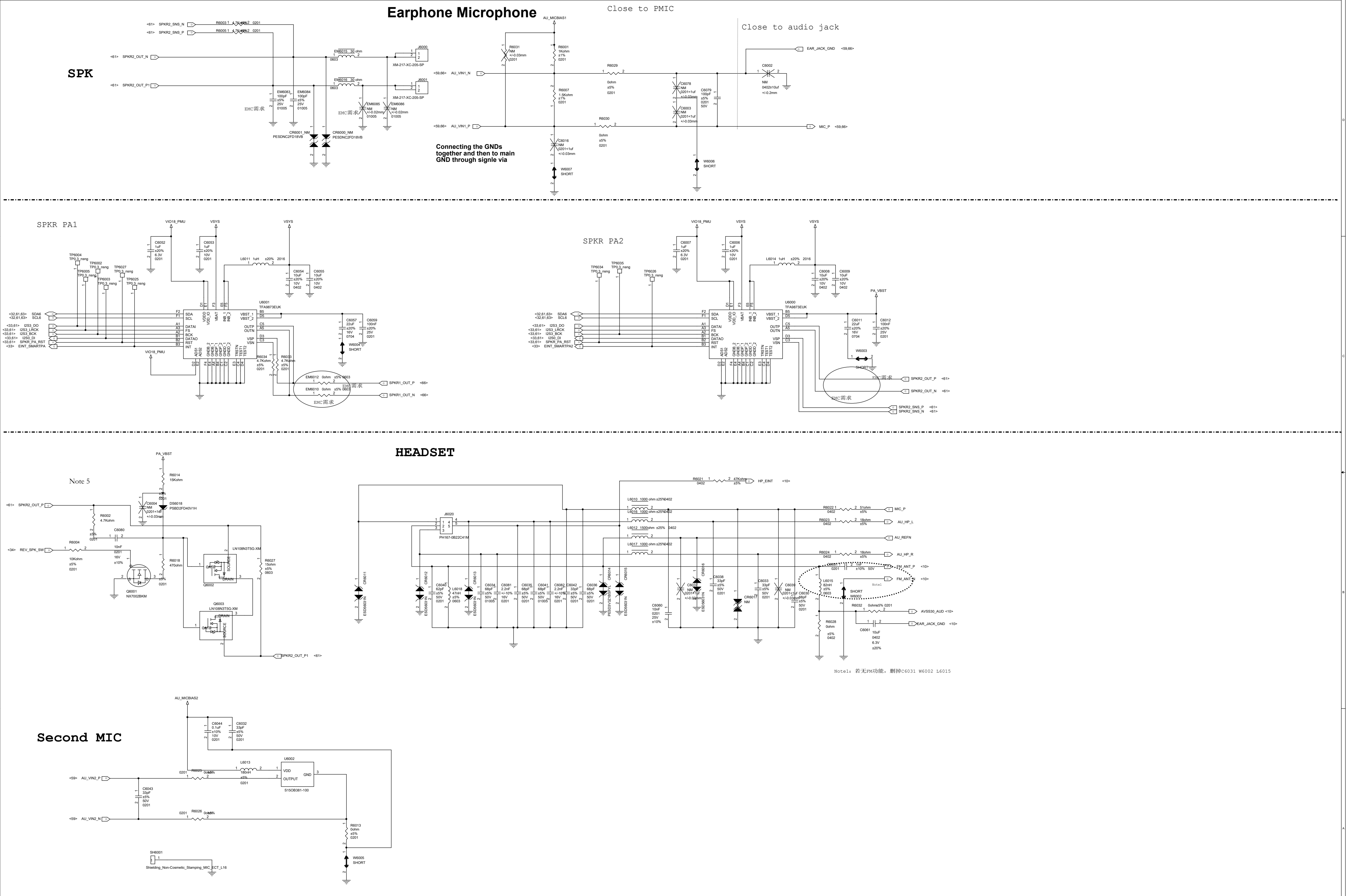
UFS 3.1



Title	Memory_1
Size A3	Document Number MEMORY
Date: Tuesday, March 14, 2023	Sheet 57 of 74

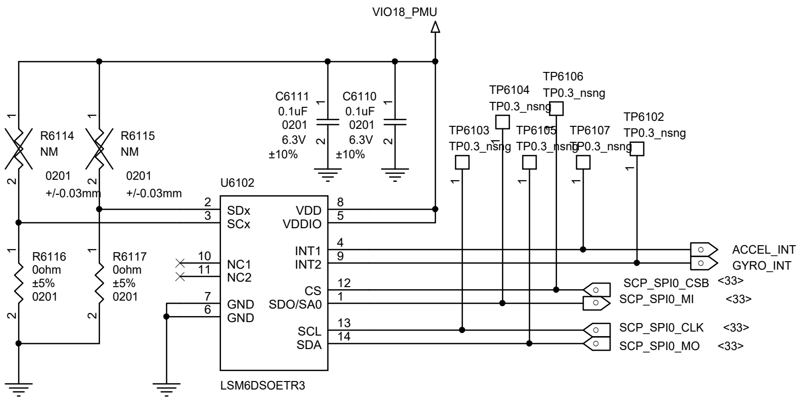


MI PHONE		
Title	MT6873_5GA	
Size A2	Document Number Reserved(for WCD)	Rev P0
Date: Tuesday, March 14, 2023	Sheet 60 of 74	

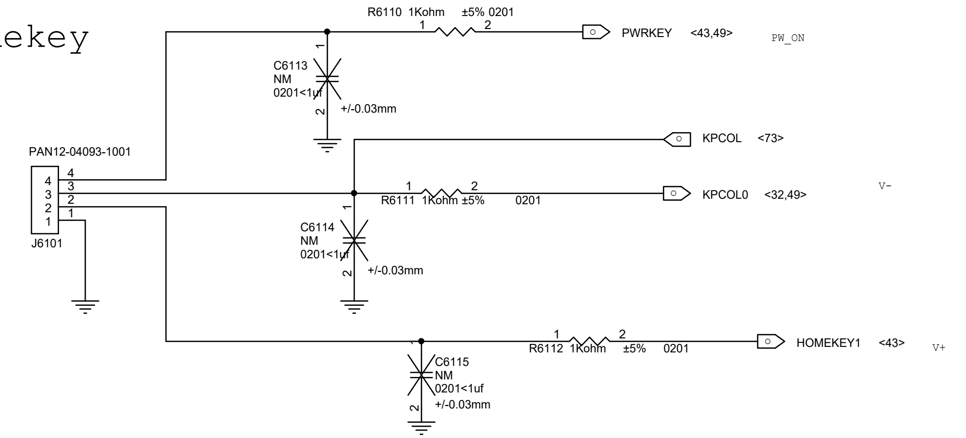


MI PHONE		
Title	MT6873_5GA	
Size A1	Document Number AUDIO	Rev
Date: Tuesday, March 14, 2023	Sheet 61 of 74	

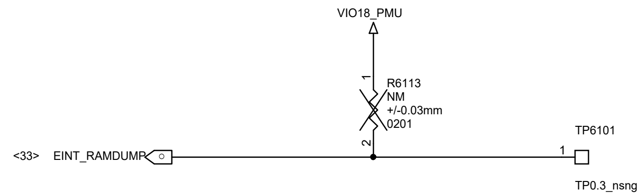
A+G



Sidekey



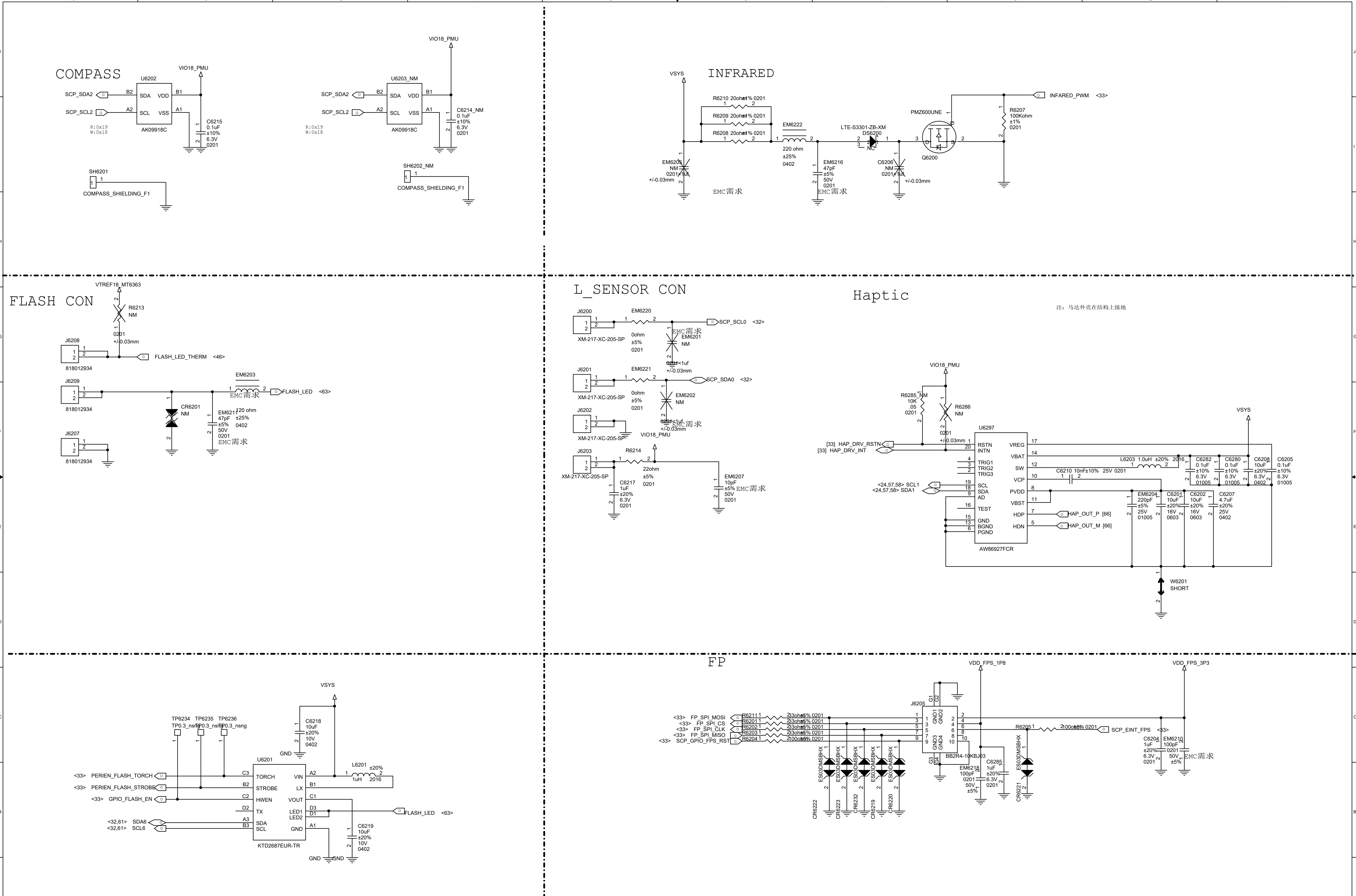
RAMDUMP debug key



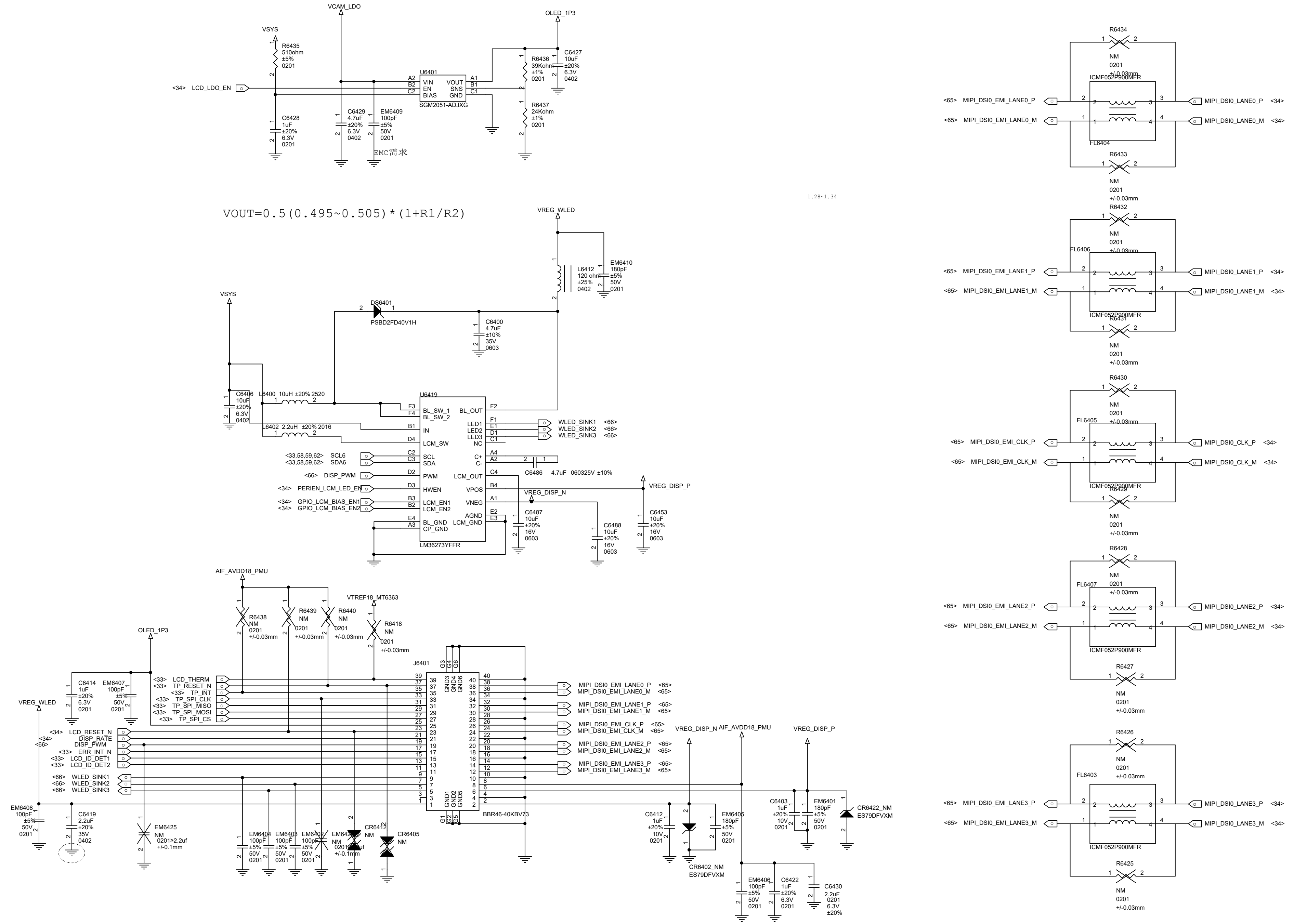
Title
J11 MAIN

Size A3 | Document Number SENSOR

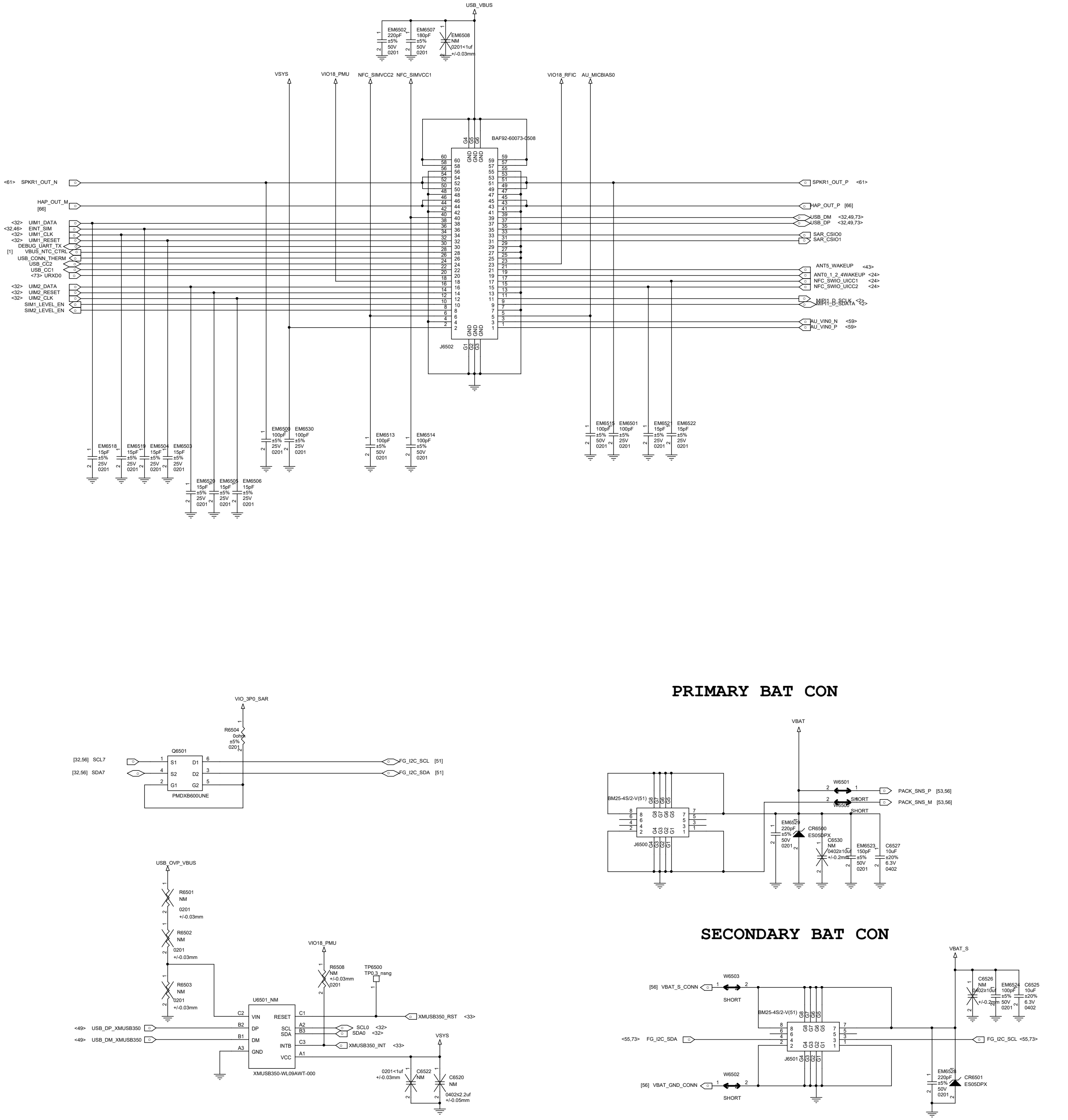
Date: Tuesday, March 14, 2023 | Sheet 62 74



MI PHONE	
Title	MT6873_5GA
Size A2	Document Number SENSOR-2	Rev P0
Date: Tuesday, March 14, 2023	Sheet 63 of 74

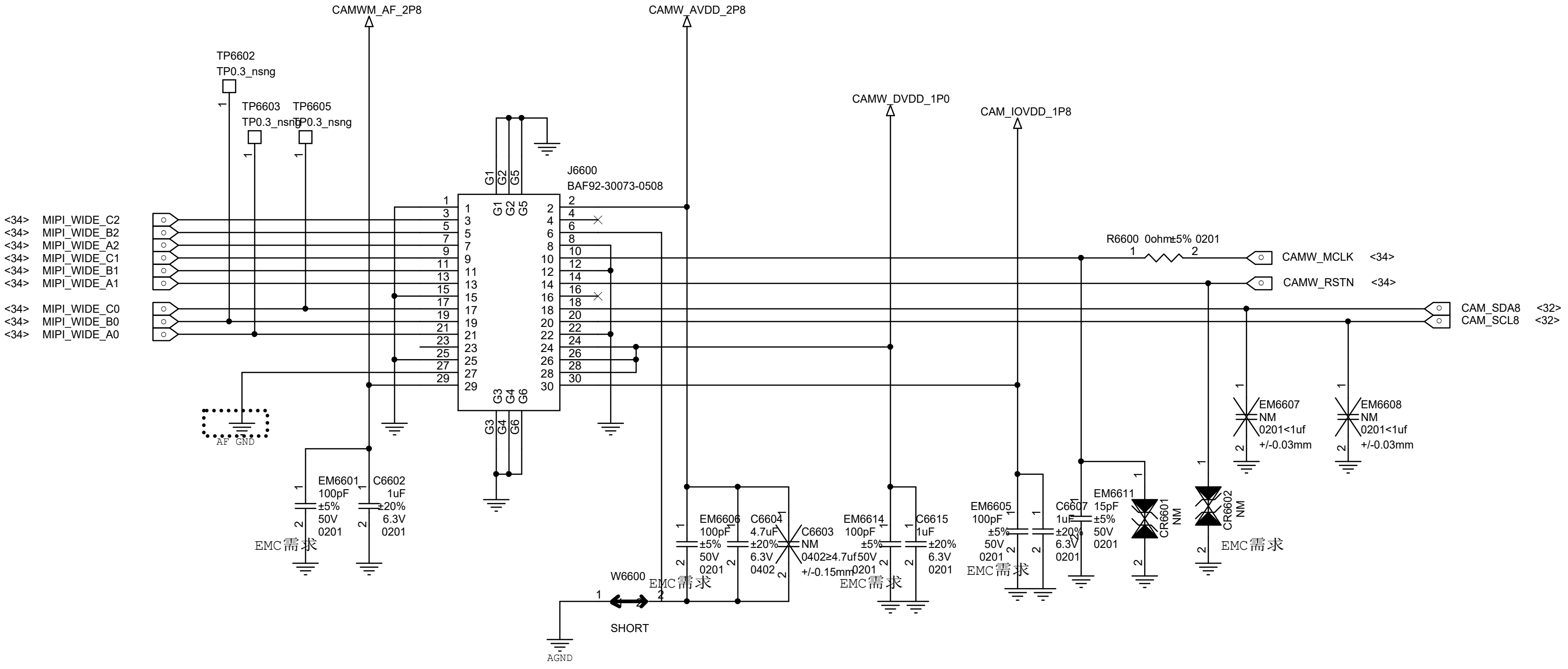


MI PHONE		
Title	MT6873_5GA	
Size A2	Document Number AMOLED	Rev P0
Date: Tuesday, March 14, 2023	Sheet 65 of 74	



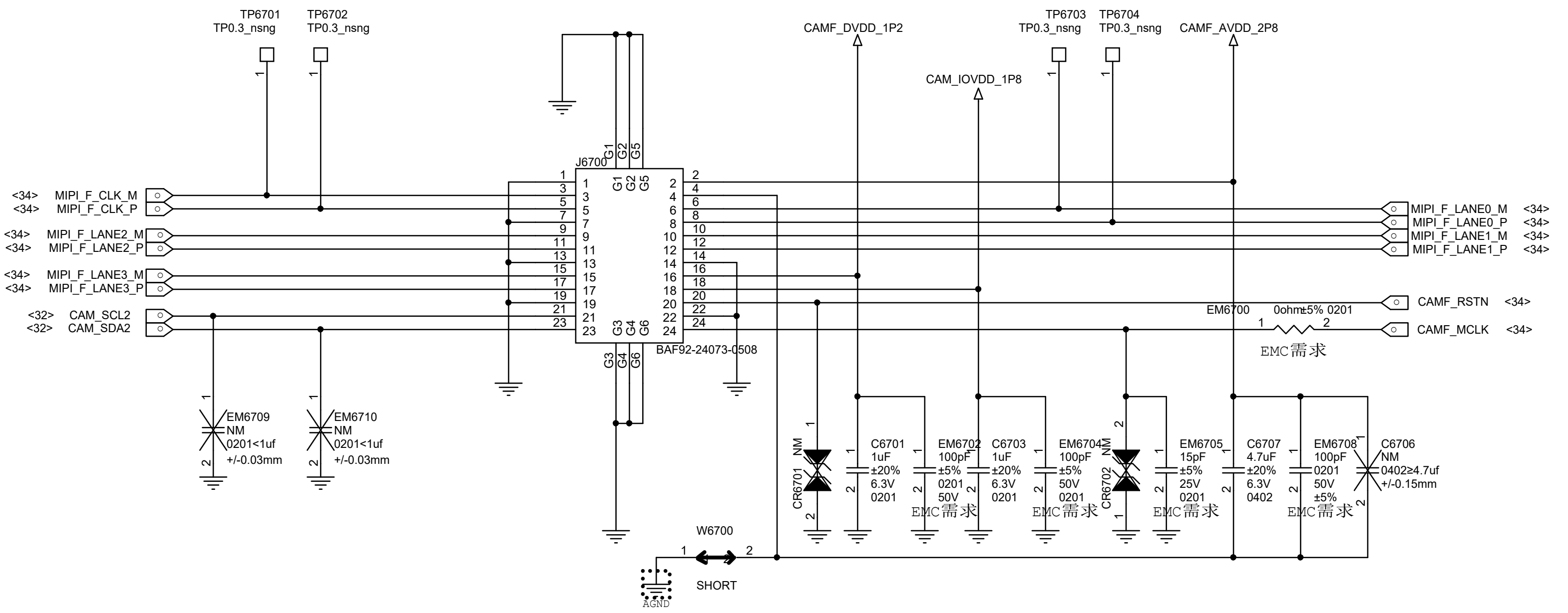
Title	J11 MAIN	
Size A1	Document Number BTB	
Date:	Tuesday, March 14, 2023	Sheet

WIDE 108M (HM2-SD) /64M(S5KGW1SD03-FGX9)



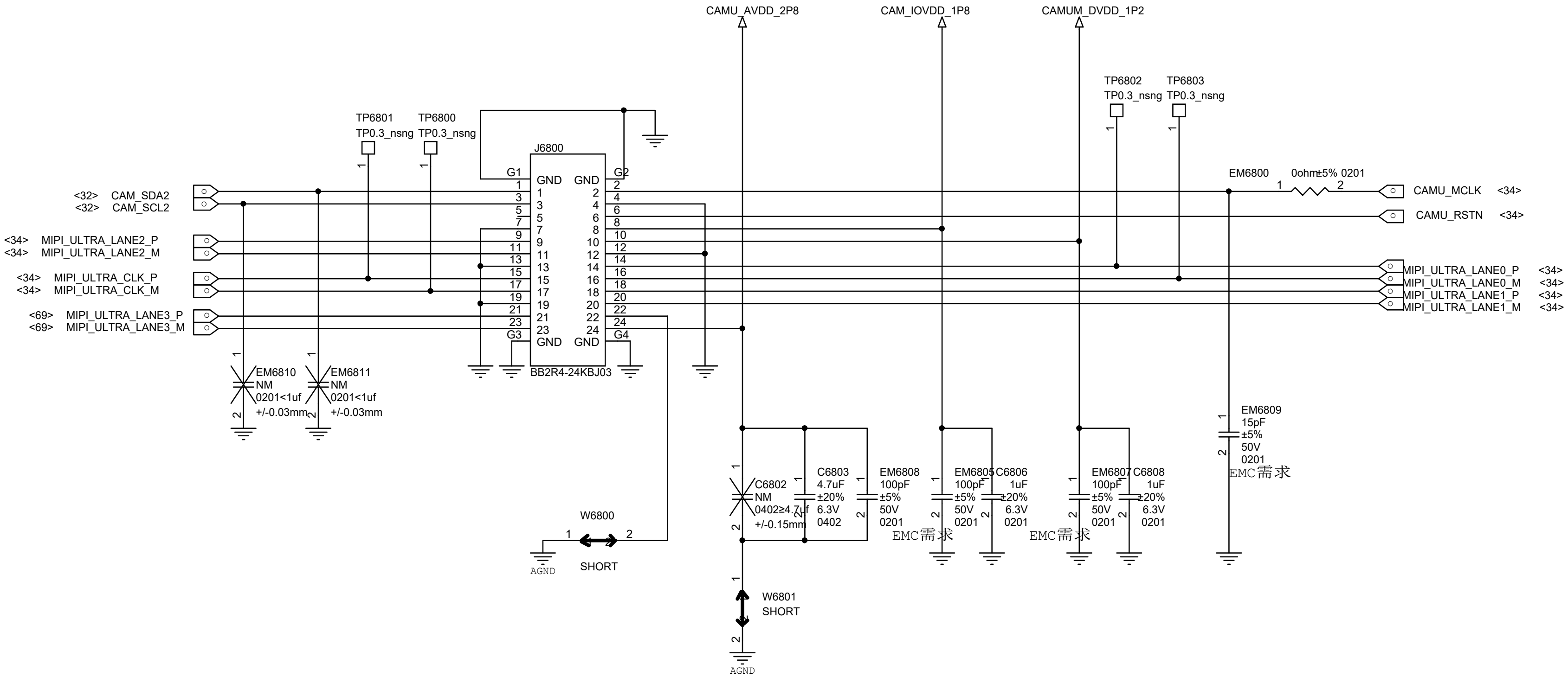
Title	J11 MAIN		
Size A3	Document Number WIDE&TELE CAMERA		
Date:	Tuesday, March 14, 2023	Sheet 67	74

front 16M OV16A1Q-GA5A



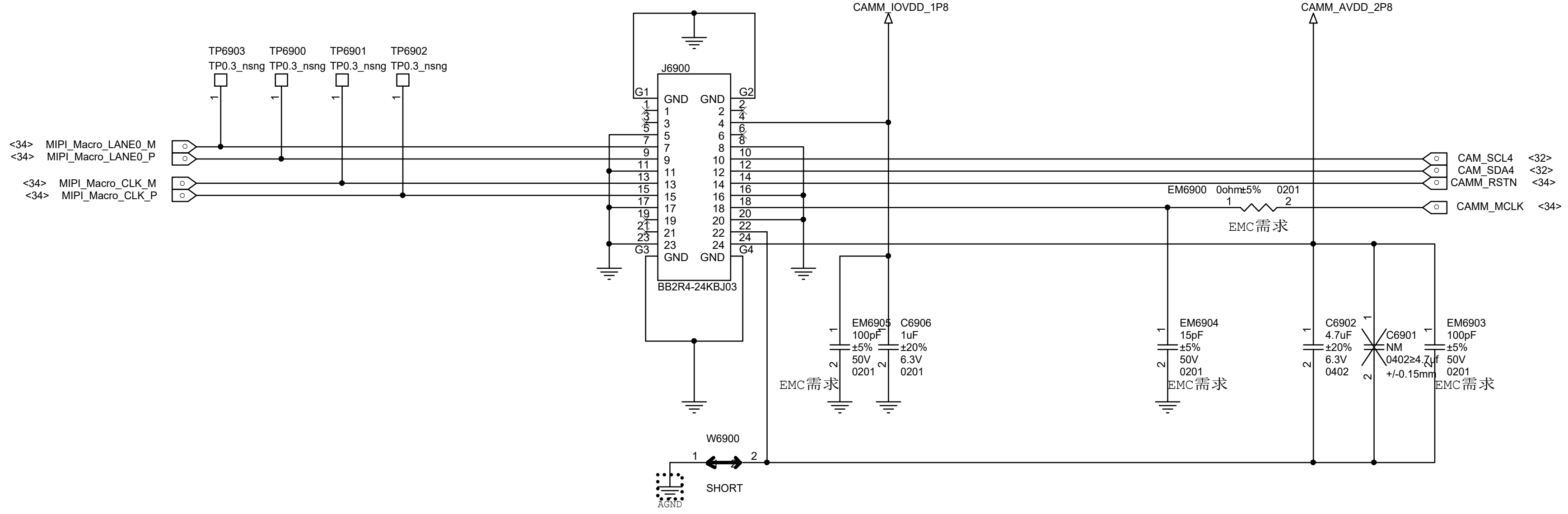
Title	J11 MAIN		
Size A3	Document Number Reserved(for Analog FM)		
Date:	Tuesday, March 14, 2023	Sheet 68	of 74

ULTRA 8M(S5K4H7)

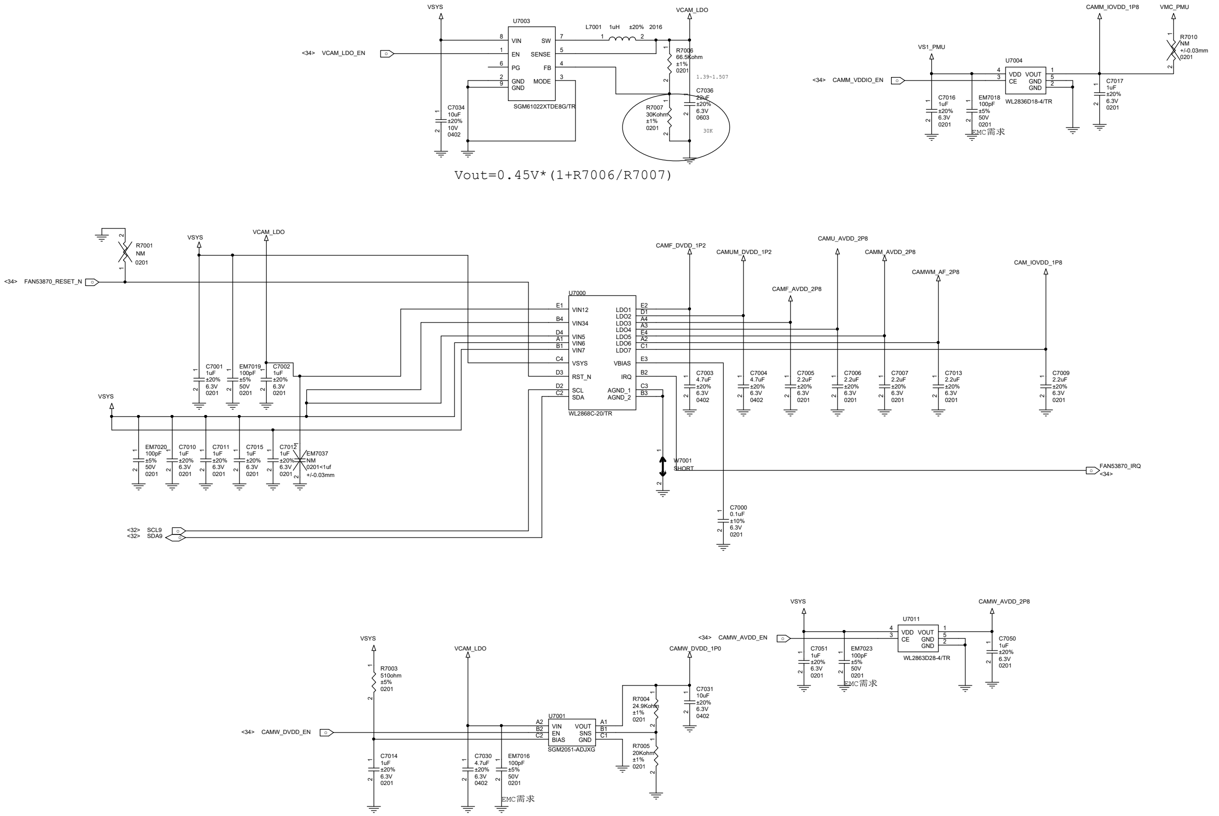


Title	J11 MAIN	
Size A3	Document Number ULTRA CAMERA	
Date: Tuesday, March 14, 2023	Sheet 69	74

MACRO 2M(OV02B10-A25A-CSP)



Title	J11 MAIN		
Size A3	Document Number ULTRA CAMERA		
Date:	Tuesday, March 14, 2023	Sheet 70	74



MI PHONE	
Title	K16_MAIN
Size A1	Document Number CAMERA LDOs
Rev	P0
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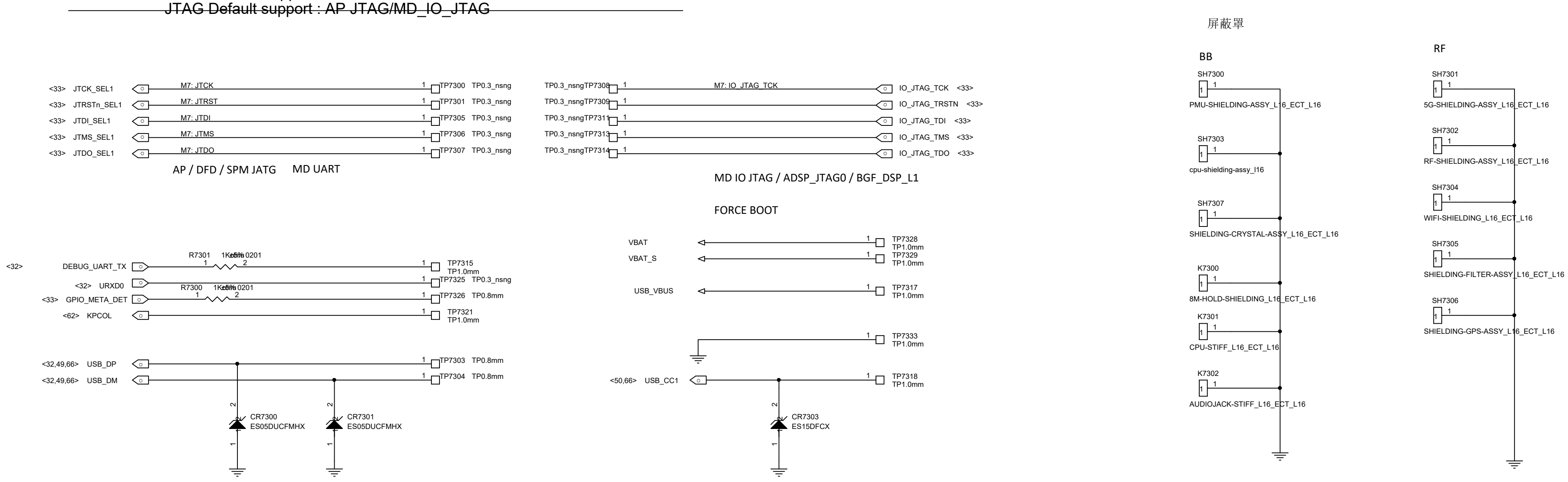
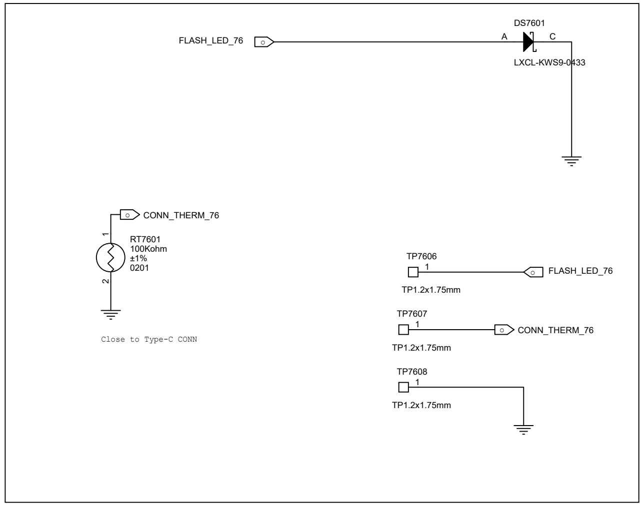
Schematic design notice of "90_DEBUG_IO" page.

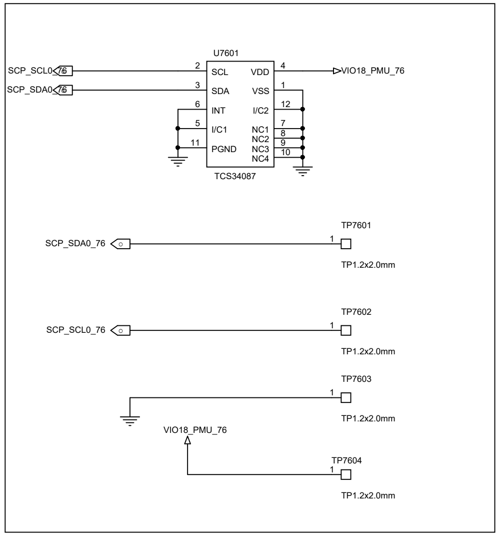
Note 90-1: UART Default support : UART0,
JTAG Default support : AP JTAG/MD_IO_JTAG



Title	J7 WLED			
Size A2	Document Number <Doc>			
Date:	Tuesday, March 14, 2023	Sheet 73	of	74

FLASH LGA





L_SENSOR LGA

Title	<Title>
Size A2	Document Number <Doc>
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